

Time Series Approaches to Cocoa Price Risk



UNIVERSITY OF
HOHENHEIM



Felix Köhler

Matriculation Number: 839611

Department of Business Mathematics and Data Science (510G)

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Abstract

This thesis asks two questions: whether Heterogeneous Autoregressive (HAR) quantile regression produces better out-of-sample distributional forecasts of cocoa futures volatility than established benchmarks, and whether augmenting the model with the El Niño–Southern Oscillation (ENSO) climate index improves long-horizon upper-tail forecasts. Daily Open-High-Low-Close prices for ICE London cocoa front-month futures spanning January 1985 to December 2025 (10,376 trading days) are used to construct a Yang-Zhang volatility proxy, validated against realized volatility from approximately one year of five-minute intraday data. Forecasts at six horizons (1 day to 12 months) are evaluated by quantile loss and QLIKE against five competing models, with Model Confidence Sets used to assess statistical significance.

On the first question, the value of direct quantile estimation is concentrated at short horizons. HAR-QR delivers the lowest upper-tail ($\tau = 0.95$) quantile loss from one day through three months, and at one day and one week it is the sole top-ranked model in a 90% Model Confidence Set that excludes historical volatility, GARCH(1,1), and the HAR-OLS pseudo-quantile. The ranking then reverses: the HAR-OLS pseudo-quantile overtakes HAR-QR at six months, and GARCH(1,1) is the best upper-tail forecaster at twelve months, 29% below HAR-QR in quantile loss, though these long-horizon differences are not statistically significant. For mean forecasts, HAR-OLS dominates at every horizon under QLIKE, MSE, and MAE. Upper-quantile coverage is near the nominal 5% at one day and one week but deteriorates as the horizon lengthens, reaching 14.6% at twelve months.

On the second question, the Niño 3.4 index worsens upper-tail quantile loss at every horizon from one to six months and improves it by only 3.3% at twelve months. Three robustness diagnostics dissolve even this residual gain: it is confined entirely to the 2022–2024 cocoa supply-crisis window—in the pre-crisis subsample ENSO augmentation *raises* twelve-month upper-tail loss by 16%—it follows a lag structure that contradicts the agronomic transmission hypothesis, and it does not replicate on Arabica coffee or raw sugar. ENSO therefore yields no robust forecast improvement at any horizon; its apparent long-horizon value is a regime-conditional, sample-specific artefact rather than a replicable structural climate channel.

The thesis makes three contributions: the first application of HAR-QR to an agricultural commodity, the first evaluation at horizons extending to twelve months, and the first empirical test of an ENSO-augmented HAR-QR specification for cocoa.

Keywords: cocoa futures, volatility forecasting, HAR model, quantile regression, ENSO, long-horizon forecasting, Yang-Zhang estimator

1 Introduction

Volatility forecasting is fundamental to financial risk management, portfolio allocation, and derivative pricing (Poon and Granger 2003). Two modeling paradigms dominate the literature. The first treats volatility as latent and inferred from returns; the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model of Bollerslev (1986) is the canonical approach. The second, enabled by high-frequency data, treats volatility as directly observable through realized measures, and the Heterogeneous Autoregressive (HAR) model of Corsi (2009) has become the dominant forecasting framework within this paradigm, capturing long-memory properties of volatility in a parsimonious, OLS-estimable structure.

The majority of volatility forecasting research focuses on conditional mean forecasts: the expected level of future volatility given current information. Yet the conditional mean provides an incomplete characterization of future volatility. When the volatility distribution is skewed or fat-tailed, as is typical for commodity markets, the conditional quantiles can diverge substantially from the mean, and understanding how the entire distribution shifts with predictor values becomes important. Quantile regression, introduced by Koenker and Bassett (1978), provides a natural framework for estimating conditional quantiles directly, without distributional assumptions.

Cocoa futures present a particularly relevant case for distributional volatility forecasting. Production is geographically concentrated in West Africa, biologically slow to adjust, and exposed to both disease and climate shocks. Between 2022 and 2024, international cocoa prices approximately tripled, an increase unprecedented in the commodity's modern trading history. Despite this salience, dedicated cocoa volatility research remains limited, and the application of HAR-based quantile regression to cocoa, or to any agricultural commodity, is an open gap in the literature. At long forecast horizons, where the autoregressive signal from lagged volatility fades, climate variables offer an additional source of predictive information. The El Niño–Southern Oscillation (ENSO) influences West African rainfall patterns with effects on cocoa yields hypothesised to materialise 6–12 months after the climate signal. Evidence from oil and broad agricultural commodity surveys suggests that ENSO can improve volatility forecasts at these longer horizons (Bouri et al. 2021; Bonato et al. 2023), motivating a test of whether this channel extends to cocoa.

This thesis addresses two research questions:

RQ1 Can HAR-based quantile regression models forecast cocoa futures volatility better than baseline models (historical volatility, GARCH, HAR-OLS) at horizons from 1 day to 12 months?

RQ2 Does the ENSO climate index improve volatility forecasts, especially at long horizons?

We address these questions by estimating HAR-QR models at multiple quantiles ($\tau \in \{0.50, 0.75, 0.95\}$) and horizons (1 day, 1 week, 1 month, 3 months, 6 months, 12 months). At each horizon, we compare out-of-sample forecast accuracy against standard benchmarks: historical rolling volatility, GARCH(1,1), and HAR estimated by ordinary least squares. For long horizons, we test whether adding ENSO improves forecasts, with particular focus on the upper quantile ($\tau = 0.95$).

The thesis makes three contributions. First, it provides the first application of HAR-QR to an agricultural commodity, extending a framework previously confined to equity and precious-metals markets. Second, it tests HAR-QR at horizons of 6–12 months, far beyond the current literature limit of approximately one month (Lyócsa and Stašek 2021). Third, it tests whether ENSO enhances quantile volatility forecasts for cocoa, a combination untested in the literature.

The remainder of the thesis is organized as follows. Section 2 reviews the literature on volatility measurement, HAR models, quantile regression in finance, and climate-commodity linkages, establishing the research gap this thesis addresses. Section 3 specifies the HAR-QR model, the multi-horizon projection approach, and the evaluation framework. Section 4 describes the ICE London cocoa futures data and validates the daily volatility proxy against realized volatility from intraday data. Section 5 presents empirical results organized by forecast horizon. Section 6 interprets findings, discusses limitations, and considers practical implications.

2 Literature Review

This section reviews the literature underlying volatility forecasting for cocoa futures. We begin with the cocoa market itself, then trace how volatility measurement evolved from simple estimators to the HAR framework. We then review quantile regression for volatility and ENSO as a climate predictor at long horizons, followed by the evaluation methods used to compare forecasts. The intersection of these strands identifies the gap this thesis addresses: no study applies HAR-based quantile regression to any agricultural commodity, and no HAR-QR study extends beyond approximately one month.

2.1 The Cocoa Market

Cocoa occupies an unusual position among agricultural commodities. Its production is geographically concentrated, biologically slow to adjust, and structurally exposed to climate variability in ways that set it apart from grains or oilseeds. These features interact to produce a market where supply disruptions are both frequent and slow to resolve, precisely the conditions under which distributional, rather than mean, volatility forecasts become most valuable.

The West African cocoa belt accounts for approximately 70% of world production, with Côte d’Ivoire and Ghana alone responsible for more than half of global output (Schroth et al. 2016; Läderach et al. 2013). Unlike grains or oilseeds, whose production is spread across multiple continents, cocoa’s concentration means that disruptions in a single region can move global prices substantially. Production is overwhelmingly smallholder-based, with approximately 90% grown on farms smaller than five hectares (Bensch et al. 2022). Since the mid-2000s, growing financial participation has added a further source of volatility (Bensch et al. 2022; Gilbert and Morgan 2010; Rogna and Tillie 2025).

A critical structural feature is the cocoa tree’s biological rigidity. New trees require five to seven years before reaching full productive capacity, with an economic lifespan of 30 to 40 years (Asante et al. 2025). Unlike annual crops, cocoa supply responds to market signals slowly, and the resulting low short-run supply elasticity amplifies price volatility whenever demand or supply shocks occur.

Cocoa is also among the most climate-sensitive perennial crops. West Africa is characterized by variable climates (Läderach et al. 2013), and drought during the critical September–December pod development period reduces yields substantially. The El Niño–Southern Oscillation (ENSO) is the primary near-term mechanism through which climate variability affects cocoa supply; Section 2.5 discusses this channel and its implications for long-horizon volatility forecasting.

The most striking recent illustration of these compounding dynamics is the 2022–



Figure 1: Ripening cocoa pods on the tree

Cocoa pods developing directly on the trunk of the cacao tree. Trees require five to seven years to reach full bearing capacity, illustrating the biological rigidity that limits short-run supply response.

Photograph taken in Ghana (2018) by Ritter Sport.



Figure 2: Smallholder post-harvest processing in Ghana

Fermented cocoa beans sun-drying on raised racks. The labour-intensive, decentralised nature of harvesting and post-harvest handling underlies the predominantly smallholder structure of West African production. Photograph taken in Ghana (2018) by Ritter Sport.

2024 price crisis, during which international cocoa prices approximately tripled (Rogna and Tillie 2025). The crisis was driven by the convergence of the Cocoa Swollen Shoot Virus Disease (CSSVD) in Ghana and adverse El Niño-related drought across Côte d’Ivoire.

Despite the salience of cocoa in global commodity markets, the volatility forecasting literature has paid it limited attention. Bensch et al. (2022) provide the most comprehensive study of cocoa price determinants, finding that fundamentals are quickly integrated into prices, but do not extend to multi-horizon forecasting or tail-risk quantification. Bergander (2025) apply LSTM neural networks with weather data but find no improvement in point forecasts, plausibly because markets anticipate weather shocks before physical effects materialize. Within broader commodity surveys, Alfeus and Nikitopoulos (2022) include cocoa among 22 markets and find that monthly volatility is the most informative component, evidence of slow-moving structural dynamics, but offer no focused analysis. No study applies HAR-based models specifically to cocoa volatility, and none has examined quantile-based tail-risk forecasting for this market.

2.2 The Evolution of Volatility Measurement

A foundational challenge in volatility research is that volatility is latent: unlike prices, which are directly observable, it must be estimated from price movements (Poon and Granger 2003). Every volatility measure is an approximation, and the choice of estimator shapes both forecast quality and evaluation validity (Patton 2011). The measurement literature has evolved through three successive approaches, each reducing the statistical noise of the volatility proxy.

The earliest volatility estimates used only closing prices. The random walk model, exponential smoothing, and rolling-window standard deviations all extrapolate past variation into the future (Poon and Granger 2003). These methods are simple and require minimal data, but the squared close-to-close return is a highly imprecise estimator of daily variance. Nonetheless, Poon and Granger (2003) review 93 studies and find that “for forecast horizons that are longer than 6 months, a simple historical method using low frequency data over a period at least as long as the forecast horizon works best,” a finding that motivates the inclusion of rolling historical volatility as a benchmark in this thesis.

A significant efficiency improvement came from exploiting the full daily price range. Parkinson (1980) demonstrated that the high-low range achieves efficiency gains of approximately five times over close-to-close methods. Rogers and Satchell (1991) extended this to allow for non-zero drift. However, both estimators assume continuous trading, an assumption violated by commodity futures markets, which experience regular overnight closures.

Yang and Zhang (2000) synthesized these approaches into an estimator that is unbiased, drift-independent, and robust to opening jumps. By estimating overnight and trading-session variance separately and combining them optimally, the Yang-Zhang estimator achieves efficiency gains of 7–8 times over close-to-close methods. For commodity markets, where overnight information frequently causes opening gaps, this robustness is particularly valuable.

The availability of intraday transaction data in the late 1990s enabled a fundamentally different approach. Andersen and Bollerslev (1998) and Andersen et al. (2001) demonstrated that realized volatility (the sum of squared intraday returns) provides a far more precise measure of daily variance, resolving much of the apparent poor performance of earlier models. However, high-frequency data are plagued by microstructure noise and are often unavailable for commodity markets over long samples. Clements et al. (2024) provide evidence that HAR models applied to daily OHLC-based volatility achieve comparable forecasting performance at medium and long horizons, validating the use of daily data when intraday records are limited, as is the case for cocoa futures over multi-decade samples.

These developments establish the Yang-Zhang estimator applied to daily OHLC data as the volatility proxy for this thesis. The following section turns from the question of what to measure to the question of how to model and forecast its temporal dynamics.

2.3 The Heterogeneous Autoregressive Model

Two modeling frameworks are central to this thesis: GARCH, which serves as the main benchmark, and the Heterogeneous Autoregressive (HAR) model, which is the primary forecasting framework. Both capture a defining feature of volatility that range-based estimators ignore: its tendency to cluster over time.

Volatility clustering was first formalized by Engle (1982), and Bollerslev (1986) generalized this into the GARCH framework, capturing both clustering and mean reversion. The GARCH(1,1) has proven remarkably difficult to beat: Hansen and Lunde (2005) evaluate 330 volatility models and conclude that “nothing beats a GARCH(1,1),” establishing it as the natural benchmark. However, GARCH cannot replicate the slow, hyperbolic decay of volatility autocorrelations observed empirically, a property known as long memory. As Corsi (2009) observes, “standard GARCH and stochastic volatility short-memory models appear like white noise once aggregated over longer time periods.”

The HAR model, introduced by Corsi (2009), addresses this limitation through a parsimonious structure that captures long memory without the estimation difficulties of fractionally integrated models. It builds on the *Heterogeneous Market Hypothesis* of Müller et al. (1997), who documented that coarse-grained volatility (e.g., weekly)

predicts fine-grained volatility (e.g., daily) more strongly than the reverse. This asymmetry reflects the presence of traders with different investment horizons: intraday speculators, swing traders, and long-term institutional investors, whose actions generate volatility that cascades across frequencies. Corsi formalized this insight into an additive model with daily, weekly, and monthly volatility components: a restricted AR(22) estimable by OLS that reproduces the autocorrelation patterns of long-memory processes.

Since its introduction, the HAR framework has been extended with jump components, realized semivariance, and time-varying coefficients (Andersen et al. 2007; Patton and Sheppard 2015; Tian et al. 2017). However, for agricultural commodities, Degiannakis et al. (2022) find that “sophisticated HAR-type models are not capable of outperforming the simple HAR in an out-of-sample exercise.” This sets a high bar for any augmentation to clear, and motivates subjecting any extension to rigorous out-of-sample testing before drawing conclusions.

For cocoa specifically, Alfeus and Nikitopoulos (2022) include it among 22 commodities in a broad HAR study, finding that “monthly volatility matters the most” for cocoa, evidence that longer-term participants exert particular influence on commodity volatility dynamics. Haugom et al. (2016) provide complementary evidence that “the effect from the monthly volatility component is strongest for all assets when predicting conditional tails.” Despite this, no study focuses specifically on cocoa volatility dynamics or applies HAR to horizons beyond one month.

2.4 Quantile Regression for Volatility

Standard volatility models, whether GARCH or HAR, estimate the conditional mean of volatility. Yet when the underlying process exhibits fat tails and regime shifts, the conditional quantiles can diverge substantially from the mean. Quantile regression, introduced by Koenker and Bassett (1978), estimates conditional quantiles directly, with each quantile having its own parameter vector. This allows the relationship between predictors and the dependent variable to differ across the distribution without imposing parametric assumptions on the error distribution.

Baur and Dimpfl (2019) demonstrate that quantile autoregression encompasses both the conditional mean and conditional variance of the return distribution through the inter-quantile range. Their approach is robust to jumps and produces variance forecasts comparable to, and in some cases more accurate than, GARCH and HAR alternatives. Haugom et al. (2016) introduced HAR-QREG (hereafter HAR-QR), combining the multi-horizon HAR structure with quantile regression, and show that the model achieves unconditional coverage rates close to nominal at substantially higher rates than historical simulation.

HAR-QR has since been applied in equity and precious-metals markets. Lyócsa and Stašek (2021) develop a complete subset quantile regression variant achieving significant improvements at horizons up to 22 trading days; Huang et al. (2022) confirm the model’s superiority using high-frequency data for Chinese indices; and Song et al. (2025) show that HAR-QR parameters improve machine learning forecasts for gold.

Despite this growing body of work, two gaps remain. First, no study applies HAR-QR to any agricultural commodity (Degiannakis et al. 2022; Bonato et al. 2023). Second, existing studies are confined to short horizons: Lyócsa and Stašek (2021) extend to 22 trading days, the longest in the literature. Whether quantile forecasts retain predictive value at horizons of 6–12 months remains untested.

2.5 Climate Variables and Commodity Volatility

At long forecast horizons, the autoregressive signal from lagged volatility fades. Climate variables offer a potential source of predictive information that operates on precisely these longer timescales.

The El Niño–Southern Oscillation (ENSO) is an irregularly periodic fluctuation in Pacific sea surface temperatures that affects weather patterns globally, cycling between warming (El Niño) and cooling (La Niña) phases every two to seven years. For cocoa, the transmission mechanism is direct: during El Niño phases, West Africa typically receives below-normal rainfall during the growing season, reducing yields in subsequent harvests. Schroth et al. (2016) document that “drought years regularly affect cocoa yields in West Africa and have particularly done so during severe El Niño years.” With the main crop harvested from October through March and the mid-crop from May through August, ENSO-driven rainfall anomalies are hypothesised to translate into production changes six to twelve months after the climate signal. We refer to this chain, from climate signal through rainfall deficit to harvest disruption and price volatility, as the *agronomic-transmission hypothesis*. The implied transmission window, if confirmed empirically, aligns with the long forecast horizons this thesis examines. Importantly, markets respond not only to realized weather but to climate expectations: Bergander (2025) observe that roughly 85% of the eventual price impact of a drought shock is reflected in futures prices before the actual crop loss occurs. This forward-looking price adjustment does not preclude predictability in *volatility*: even when the expected future price level rapidly impounds climate information, the conditional dispersion of realized outcomes may remain systematically forecastable from the ENSO state.

At the commodity price level, Ubilava (2018) examines ENSO–price relationships across 43 commodities, finding significant effects for tropically grown agricultural products including cocoa, with “more amplified price responses during El Niño events.” Recent work has extended this analysis from price levels to volatility. Bouri et al. (2021)

find that ENSO improves oil volatility forecasts at horizons of 2–4 years, far beyond where lagged volatility alone provides value. Bonato et al. (2023) extend this to 16 agricultural commodities, finding that “the evidence of predictive value of El Niño and La Niña events strengthens at the longer term forecast horizons,” precisely where standard autoregressive models lose power. More broadly, Guo et al. (2025) demonstrate that climate risk indices significantly improve commodity volatility forecasts, with models incorporating climate risk “outperforming traditional ones in economic value.”

These findings establish ENSO as a physically motivated, exogenous predictor with demonstrated value for oil and agricultural commodity volatility at long horizons, but whether this extends to cocoa remains untested, and no prior study has combined ENSO with quantile regression.

2.6 Evaluating Volatility Forecasts

A distinctive challenge in evaluating volatility forecasts is that the variable of interest is unobservable, even *ex post* (Patton 2011). Every volatility measure is an imperfect proxy, and noise in the proxy can produce misleading model rankings depending on the loss function used.

Patton (2011) derives conditions for loss functions that yield rankings invariant to the choice of volatility proxy. Among common loss functions, only mean squared error (MSE) and QLIKE satisfy these conditions. Non-robust loss functions can produce rankings that deviate by factors of one quarter to three, explaining widespread conflicting findings across studies. Between the two robust measures, QLIKE penalizes under-prediction proportionally more heavily, making it suitable when the forecasting goal involves risk assessment (Lyócsa and Stašek 2021; Clements et al. 2024; Ghysels and Marcellino 2018). Mean absolute error (MAE) is a further robustness check that captures deviations in the volatility level rather than its square.

Hansen et al. (2011) propose the Model Confidence Set (MCS) for simultaneous comparison of multiple models, identifying the set $\mathcal{M}^*$ containing the best model at a given confidence level. The MCS “acknowledges the limitations of the data”: informative samples yield small confidence sets, while uninformative samples include many models, particularly relevant when evaluation noise from an unobservable target reduces the power to discriminate (Clements et al. 2024; Degiannakis et al. 2022).

For quantile forecasts, Christoffersen (1998) establishes conditional coverage testing: a well-specified quantile forecast at level τ should produce violations at rate τ , and these violations should be serially independent. The quantile loss function (the check function used in estimation) also serves as a natural evaluation metric, penalizing over- and under-prediction asymmetrically according to τ (Lyócsa and Stašek 2021). Puke and Schweikert (2026) show that aligning the estimation and evaluation loss function

produces significant forecast gains; since HAR-QR uses the check function for both estimation and evaluation, it is inherently coherent by construction.

3 Methodology

This section specifies the econometric framework for forecasting cocoa futures volatility at horizons from one day to twelve months. We first define the volatility measure, then specify the HAR-QR model and its ENSO-augmented extension, describe the benchmark models, and finally lay out the evaluation framework. Table 1 provides an at-a-glance summary of the five models before the detailed specifications follow.

Table 1: Overview of models compared in this study, ordered from simplest to most complex.

Feature	Benchmarks			Main models	
	Hist. Vol.	GARCH(1,1)	HAR-OLS	HAR-QR	HAR-X-QR
Model class	Non-param.	ARCH-type	Linear reg.	Quantile reg.	Quantile reg.
HAR lags ($d/w/m$)	No	No	Yes	Yes	Yes
ENSO regressor	No	No	No	No	Yes
Quantile output	Empirical	Closed-form	None	Direct	Direct
Estimation	—	MLE	OLS	Check fn.	Check fn.
Multi-period design	Rolling window	Recursive	Direct	Direct	Direct
Thesis role	Naive BM	Param. BM	HAR benchmark	Main (RQ1)	Main (RQ2)

Direct = horizon-specific model estimated by direct projection; Recursive = closed-form GARCH mean recursion, quantiles via Gaussian closed-form (Eq. 13); Check fn. = check-function (quantile) loss; BM = benchmark; Hist. Vol. = rolling 252-day window with empirical quantiles.

3.1 Volatility Measure

We measure daily volatility using the Yang-Zhang estimator, which combines overnight and trading-session variance components:

$$\hat{\sigma}_{YZ}^2 = \hat{\sigma}_O^2 + k\hat{\sigma}_C^2 + (1 - k)\hat{\sigma}_{RS}^2, \quad (1)$$

where $\hat{\sigma}_O^2$ captures overnight variance, $\hat{\sigma}_C^2$ captures open-to-close variance, and $\hat{\sigma}_{RS}^2$ is the Rogers-Satchell component that exploits the daily range. The weighting factor $k = 0.34/(1.34 + (n + 1)/(n - 1))$ minimizes estimator variance.

This choice is motivated by two considerations. First, Yang-Zhang achieves efficiency gains over close-to-close estimators and second, it can be computed over our full 40-year daily sample, unlike realized volatility which requires high-frequency data. We validate the Yang-Zhang proxy against five-minute realized volatility in Section 4, comparing rolling-window aggregates (22-day averages) of both measures via a Mincer-Zarnowitz regression. This comparison at the multi-day level aligns with the HAR model’s use of averaged volatility as both predictors and targets.

3.2 The HAR-QR Model

Combining the HAR framework with quantile regression yields the HAR-QR model. The HAR-QR model for horizon h and quantile τ is

$$Q_\tau(\bar{\sigma}_{t:t+h} \mid \mathcal{F}_t) = \beta_0(\tau) + \beta_d(\tau)\sigma_t^{(d)} + \beta_w(\tau)\sigma_t^{(w)} + \beta_m(\tau)\sigma_t^{(m)}, \quad (2)$$

where $\mathcal{F}_t$ denotes the information set at time t . The volatility components capture variation at three frequencies:

$$\sigma_t^{(d)} = \sigma_t, \quad (3)$$

$$\sigma_t^{(w)} = \frac{1}{5} \sum_{i=0}^4 \sigma_{t-i}, \quad (4)$$

$$\sigma_t^{(m)} = \frac{1}{22} \sum_{i=0}^{21} \sigma_{t-i}. \quad (5)$$

The key feature of quantile regression is that each quantile τ has its own coefficient vector, allowing the relationship between predictors and future volatility to differ across the distribution. For the median ($\tau = 0.50$), the coefficients may resemble those from OLS estimation. For the upper quantile ($\tau = 0.95$), the coefficients capture how the predictors relate to volatility during high-volatility episodes. We report three quantiles: $\tau \in \{0.50, 0.75, 0.95\}$. The median $\tau = 0.50$ serves as an anchor to OLS-based point forecasts. The intermediate quantile $\tau = 0.75$ traces the transition from central to tail behaviour. The upper quantile $\tau = 0.95$ is of primary interest, as it captures the conditional upper bound of volatility, the regime where mean forecasts are most likely to understate actual outcomes. Lower quantiles ($\tau = 0.05, 0.25$) are omitted as the thesis focuses on upper-tail risk.

The dependent variable is the average daily volatility over the forecast horizon, following the HAR literature convention (Ghysels and Marcellino 2018; Clements et al. 2024):

$$\bar{\sigma}_{t:t+h} = \frac{1}{h} \sum_{i=1}^h \sigma_{t+i}. \quad (6)$$

This averaging ensures comparability across horizons and aligns with the HAR model's construction, where weekly and monthly components are themselves averages of daily volatility.

We adopt the *direct* forecasting approach: a separate model is estimated for each horizon h , using $\bar{\sigma}_{t:t+h}$ as the dependent variable. This contrasts with *iterated* forecasting, which estimates a one-step model and iterates forward. Ghysels and Marcellino (2018) find that direct forecasting is preferred for realized volatility models. For quantile regression, direct estimation is particularly natural, as it produces the conditional

quantile at horizon h without the distributional assumptions required to iterate quantile forecasts. Our longest horizons (6–12 months) would require iterating hundreds of daily steps, amplifying any model misspecification.

We estimate HAR-QR at six horizons: $h \in \{1, 5, 22, 66, 132, 264\}$ trading days, corresponding to 1 day, 1 week, 1 month, 3 months, 6 months, and 12 months. The short horizons (1 day to 1 month) serve to validate the methodology against existing literature: HAR-QR has been tested at horizons up to 22 trading days (Lyócsa and Stašek 2021). The long horizons (6 and 12 months) constitute the core contribution, testing whether HAR-QR can produce useful forecasts at horizons where no prior study has ventured.

At long horizons, the autoregressive signal from lagged volatility fades. We therefore augment the HAR-QR model with the ENSO climate index:

$$Q_\tau(\bar{\sigma}_{t:t+h} \mid \mathcal{F}_t) = \beta_0(\tau) + \beta_d(\tau)\sigma_t^{(d)} + \beta_w(\tau)\sigma_t^{(w)} + \beta_m(\tau)\sigma_t^{(m)} + \gamma(\tau)\text{ENSO}_t, \quad (7)$$

We refer to this augmented specification as the *HAR-X-QR* model, where the “-X” suffix denotes the inclusion of the exogenous ENSO regressor. Here ENSO_t is the Niño 3.4 sea surface temperature anomaly index from NOAA, measured monthly. The coefficient $\gamma(\tau)$ captures the marginal effect of ENSO conditions on future volatility at quantile τ .

We use *contemporary* ENSO, meaning ENSO_t predicts $\bar{\sigma}_{t:t+h}$, rather than explicitly lagged ENSO values. This specification is standard in the climate-commodity literature (Bouri et al. 2021; Bonato et al. 2023) and is causal: the Niño 3.4 anomaly for month t is publicly observable before any forecast is issued, so no forward-looking information enters the model. The sensitivity of this specification to alternative ENSO lags is examined in Section 5.5.

Positive Niño 3.4 anomalies indicate El Niño conditions; negative anomalies indicate La Niña. Full details of the index construction, sample coverage, and monthly-to-daily alignment are provided in Section 4.2.

The HAR-QR model is estimated by minimizing the check function loss:

$$\hat{\beta}(\tau) = \arg \min_{\beta} \sum_{t=1}^{T-h} \rho_\tau(\bar{\sigma}_{t:t+h} - X_t' \beta), \quad (8)$$

where $X_t = (1, \sigma_t^{(d)}, \sigma_t^{(w)}, \sigma_t^{(m)})'$ for the basic model, or $X_t = (1, \sigma_t^{(d)}, \sigma_t^{(w)}, \sigma_t^{(m)}, \text{ENSO}_t)'$ for the extended model. The check function is defined as:

$$\rho_\tau(u) = u \cdot (\tau - \mathbf{1}_{u < 0}) = \begin{cases} \tau \cdot u & \text{if } u \geq 0, \\ (\tau - 1) \cdot u & \text{if } u < 0. \end{cases} \quad (9)$$

This asymmetric loss function penalizes under-prediction more heavily when τ is large and over-prediction more heavily when τ is small, ensuring that the estimated quantile achieves the target coverage rate. The minimization is a linear programming problem solved efficiently using the simplex algorithm (Koenker and d'Orey 1987).

3.3 Benchmark Models

We compare HAR-QR and HAR-X-QR against three benchmarks that span the major modeling paradigms discussed in the literature review.

The first benchmark is *rolling historical volatility* with a one-year window ($w = 252$):

$$\hat{\sigma}_{t+h}^{\text{hist}} = \sqrt{\frac{1}{w} \sum_{i=1}^w (r_{t-i+1} - \bar{r})^2}. \quad (10)$$

For quantile forecasts, we compute empirical quantiles of the historical volatility distribution: sort the w past volatilities and select the value at position $\lceil \tau \cdot w \rceil$. This model captures only unconditional variation and serves as a naive baseline.

The second benchmark is the *GARCH(1,1)* model (Bollerslev 1986):

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \quad (11)$$

For mean forecasts at horizon h , the GARCH(1,1) produces closed-form predictions:

$$\mathbb{E}_t[\sigma_{t+h}^2] = \bar{\sigma}^2 + (\alpha + \beta)^{h-1}(\sigma_{t+1}^2 - \bar{\sigma}^2), \quad (12)$$

where $\bar{\sigma}^2 = \omega/(1 - \alpha - \beta)$ is the unconditional variance and $(\alpha + \beta)^{h-1}$ governs the speed of mean reversion. At long horizons, all h -step forecasts converge to $\bar{\sigma}^2$, irrespective of current conditions, a structural limitation of daily GARCH at 6–12 month horizons.

For quantile forecasts we adopt a closed-form Gaussian approach consistent with the normal-innovation assumption used in estimation. The mean forecast $\hat{\sigma}_{t,h}$ is obtained as the square root of the average h -step variance, $\hat{\sigma}_{t,h} = \sqrt{h^{-1} \sum_{k=1}^h \mathbb{E}_t[\sigma_{t+k}^2]}$. Forecast uncertainty is captured by the in-sample standard deviation s_t of daily volatility over the estimation window. The τ -quantile forecast is then

$$\hat{Q}_{\tau,t}^{\text{GARCH}} = \hat{\sigma}_{t,h} + \Phi^{-1}(\tau) s_t, \quad (13)$$

where $\Phi^{-1}(\tau)$ is the standard normal quantile function.

The third benchmark is *HAR-OLS*, the standard HAR model estimated by ordinary least squares. OLS yields only a conditional mean forecast $\hat{\mu}_{t,h}$, but a distributional forecast can be constructed from the in-sample residuals. Let $\hat{\varepsilon}_i = \sigma_i - \hat{\mu}_i$ denote the

residuals from the training window. The τ -quantile forecast is then

$$\hat{Q}_{\tau,t}^{\text{OLS}} = \hat{\mu}_{t,h} + \hat{F}_{\varepsilon}^{-1}(\tau), \quad (14)$$

where $\hat{F}_{\varepsilon}^{-1}(\tau)$ is the empirical τ -quantile of the training-window residuals. This pseudo-quantile approach requires no distributional assumption and produces forecasts that are directly comparable to HAR-QR: if HAR-QR provides no improvement over HAR-OLS at quantile τ , then conditioning on the HAR structure suffices and quantile regression adds no value. Comparing the two at $\tau = 0.50$ tests symmetry of the conditional distribution; comparing at $\tau = 0.95$ tests whether the upper tail is better captured by explicit quantile estimation.

3.4 Evaluation Framework

Evaluating volatility forecasts is complicated by the fact that volatility is unobserved, and every evaluation compares forecasts against an imperfect proxy (Patton 2011). We evaluate forecasts against Yang-Zhang volatility computed from realized prices, using loss functions that Patton (2011) has shown to be robust to noise in the volatility proxy.

For mean and median forecasts, we use QLIKE as the primary metric. All five models are therefore compared on QLIKE: the three benchmarks contribute their mean forecasts, while HAR-QR and HAR-X-QR contribute their median ($\tau = 0.50$) forecasts, which provide a natural reference point for comparison with OLS-based point forecasts.

$$\text{QLIKE} = \frac{1}{T} \sum_{t=1}^T \left(\log \hat{\sigma}_t^2 + \frac{\sigma_t^2}{\hat{\sigma}_t^2} \right). \quad (15)$$

Patton (2011) derives necessary and sufficient conditions for the loss function to yield forecast rankings that are robust to proxy noise, showing that among nine commonly used loss functions, only MSE and QLIKE satisfy these conditions. We prefer QLIKE because it penalizes under-prediction of volatility proportionally more than MSE, which is appropriate when under-prediction is costlier than over-prediction, a property particularly relevant for upper-quantile forecasts. As Puke and Schweikert (2026) demonstrate, aligning estimation and evaluation loss functions, what they call *coherent forecasting*, yields significant accuracy gains. Our HAR-QR models are coherent by construction: the check function (Equation 9) serves as both the estimation objective and the evaluation metric.

As a robustness check, we additionally report mean squared error (MSE) and mean

absolute error (MAE) for all point-forecast models:

$$\text{MSE} = \frac{1}{T} \sum_{t=1}^T (\sigma_t - \hat{\sigma}_t)^2, \quad (16)$$

$$\text{MAE} = \frac{1}{T} \sum_{t=1}^T |\sigma_t - \hat{\sigma}_t|. \quad (17)$$

MSE penalizes large errors quadratically and is a member of the proxy-robust class alongside QLIKE (Patton 2011); including it confirms that QLIKE rankings are not sensitive to the choice of loss function within that class. MAE penalizes all deviations linearly and is not proxy-robust, but provides an interpretable measure of average forecast error in the original volatility units.

For quantile forecasts, we use quantile loss:

$$\text{QL}_\tau = \frac{1}{T} \sum_{t=1}^T \rho_\tau(\sigma_t - \hat{Q}_{\tau,t}). \quad (18)$$

As a complementary diagnostic, we assess the calibration of quantile forecasts through their *empirical violation rate*. For a given quantile level τ , the violation rate measures the fraction of out-of-sample observations that exceed the quantile forecast:

$$\widehat{VR}_\tau = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{\sigma_t > \hat{Q}_{\tau,t}\}, \quad (19)$$

where $\mathbf{1}\{\cdot\}$ is the indicator function. A well-calibrated τ -quantile forecast should satisfy $\widehat{VR}_\tau \approx 1 - \tau$. For example, the $\tau = 0.95$ quantile should be exceeded approximately 5% of the time; systematic departures indicate miscalibration. Violation rates above $1 - \tau$ signal that the model underestimates the quantile (too narrow), while rates below $1 - \tau$ indicate overestimation (too conservative). This diagnostic complements quantile loss, which measures average forecast accuracy but does not directly reveal whether the nominal coverage level is achieved.

Calibration is tested formally with the likelihood-ratio framework of Christoffersen (1998). The unconditional coverage test (LR_{uc}, χ_1^2) tests whether the violation frequency equals the nominal rate $1 - \tau$; the independence test (LR_{ind}, χ_1^2) tests whether violations arrive independently rather than in clusters, using a first-order Markov alternative; and the conditional coverage test ($LR_{cc} = LR_{uc} + LR_{ind}, \chi_2^2$) combines both properties. One complication arises from the daily rolling design: at horizons $h > 1$, consecutive forecasts target overlapping h -day windows, so consecutive violation indicators are dependent by construction and the independence test would reject mechanically. We therefore compute the test statistics on the non-overlapping subsequence of every h -th forecast, while reporting the empirical violation rate over all

forecasts. At the longest horizons this leaves few non-overlapping observations (28 at twelve months), so the tests lose power and the raw violation rate becomes the more informative diagnostic.

To determine whether differences in forecast accuracy are statistically significant, we use the Model Confidence Set (MCS) of Hansen et al. (2011). The MCS identifies the set of models $\mathcal{M}^*$ that contains the best model with a given confidence level, avoiding the need to specify a single benchmark. Starting from the full collection of competing models, the procedure iteratively tests for equal predictive ability and eliminates the worst-performing model until non-rejection occurs. We implement MCS with 10,000 stationary bootstrap replications using the max- t statistic; models in $\mathcal{M}^*$ at the 90% confidence level are considered statistically equivalent.

The forecast evaluation is conducted out-of-sample using a rolling estimation window of 2,520 trading days (approximately 10 years). At each forecast origin t , we estimate parameters on the 2,520 most recent observations whose h -day target windows are fully realized by t — feature dates $s \in \{t - h - 2519, \dots, t - h\}$, so that no training target extends beyond the forecast origin — generate forecasts, and roll forward by one day. The window rolls *daily* to maximize the number of forecast–realization pairs for statistical evaluation. With approximately 30 years of out-of-sample data, this yields thousands of pairs for statistical testing.

Several assumptions underpin the methodology. First, we assume *local stationarity*: volatility dynamics are stable within each estimation window, though they may shift across windows. The rolling-window design accommodates gradual parameter changes. Second, we assume that the Yang-Zhang estimator provides a reasonable proxy for latent volatility, which we validate empirically in Section 4. Third, we maintain *linear conditional quantiles* for parsimony, acknowledging that non-linear extensions could potentially improve forecasts. Fourth, we assume ENSO is exogenous to the cocoa market, which is physically reasonable.

Based on the literature, we expect that at short horizons (1 day to 1 month), HAR-QR will match or outperform benchmarks. At long horizons (6–12 months), standard HAR-QR will lose predictive power; we test whether HAR-X-QR augmented with ENSO recovers predictive value, examining whether climate information provides incremental value precisely where the autoregressive signal fades.

Should HAR-X-QR outperform HAR-QR at long horizons, a single full-sample comparison cannot by itself establish whether the gain reflects a structural climate channel or a sample-specific coincidence. We therefore subject any ENSO improvement to three diagnostics, reported in Section 5.5. First, a *temporal-stability* check traces the cumulative ENSO gain as the evaluation horizon advances through time using an expanding-window diagnostic. For each calendar month T in the out-of-sample period, the quantile loss difference $\Delta\text{QL}(T) = \text{QL}_{\text{HAR-X-QR}}(\{t \leq T\}) - \text{QL}_{\text{HAR-QR}}(\{t \leq T\})$

is computed over all forecast–realisation pairs up to and including T . A structural ENSO–volatility channel should produce a steadily accumulating gain throughout the evaluation period, not a gain concentrated in a single episode. Second, a *lag-structure* check re-estimates HAR-X-QR with the Niño 3.4 index lagged 0, 1, 3, 6, and 12 months. Under the agronomic-transmission hypothesis the predictive content should strengthen at lags of 6–12 months, matching the window over which ENSO-driven rainfall anomalies are hypothesised to affect harvests. This is consistent with the broader finding in the literature that ENSO’s predictive value for commodity volatility concentrates at longer horizons: Bouri et al. (2021) document gains at 2–4 year horizons for oil-price realized volatility, while Bonato et al. (2023) find a general tendency for ENSO predictability to strengthen at longer forecast horizons across 16 agricultural commodities. Third, a *cross-commodity* check applies the identical HAR-X-QR specification to Arabica coffee and raw sugar, two tropical soft commodities that share weather exposure and downstream markets with cocoa (Ge et al. 2025), since a genuine climate channel should produce comparable gains across commodities grown in the same equatorial belt (Ubilava 2018; Bonato et al. 2023). Because continuous futures price series for coffee and sugar are available from approximately 2000 onward, the out-of-sample evaluation windows for these commodities span roughly 15 years — approximately half the cocoa sample. The cross-commodity results should be interpreted in light of this shorter evaluation period.

4 Data

This section describes the data used to estimate and evaluate the volatility forecasting models developed in Section 3. Two primary datasets are required: daily futures prices for ICE London cocoa contracts and the Niño 3.4 climate index measuring El Niño–Southern Oscillation (ENSO) activity. After describing the cocoa futures data, including contract specifications, the construction of a continuous price series, and the variables derived from it, we turn to the ENSO index used as an exogenous predictor at long horizons. The section concludes with descriptive statistics documenting the stylized facts of cocoa returns and volatility.

4.1 Cocoa Futures Data

The primary data consist of daily Open-High-Low-Close (OHLC) prices and trading volume for ICE Futures Europe cocoa futures contracts. ICE London cocoa futures are standardized contracts for the delivery of 10 metric tonnes of cocoa beans. Each contract specifies delivery in one of five months: March, May, July, September, or December. Prices are quoted in British pounds sterling (GBP) per metric tonne, and trading occurs on the IntercontinentalExchange (ICE) electronic platform during European market hours. The data were obtained from Refinitiv Workspace via the corporate license held by Ritter Sport GmbH & Co. KG.

Futures contracts have finite maturity, requiring the construction of a continuous price series for volatility analysis. We use the generic front-month continuous contract, denoted LCCc1 in the Refinitiv system. This series rolls to the next available contract approximately one month before expiration, maintaining exposure to the most liquid contract while avoiding delivery. The front-month contract typically exhibits the highest trading volume and the tightest bid-ask spreads, making it the standard choice for volatility research on commodity futures (Degiannakis et al. 2022; Alfeus and Nikitopoulos 2022). Rolling between contracts creates artificial price jumps when the new contract trades at a different level than the expiring one (the roll yield). For volatility estimation using OHLC data, these jumps are largely captured within the daily range and do not require explicit adjustment. The Yang-Zhang estimator (Section 3.1) is specifically designed to be robust to opening jumps, which include both overnight news and contract roll effects.

The daily dataset spans 10,376 trading days from January 2, 1985, to December 15, 2025. This extended sample offers several advantages. It encompasses multiple El Niño and La Niña cycles, providing sufficient variation in the ENSO predictor for long-horizon models. It includes diverse market regimes: the long period of relative stability from 1990–2010, the financialization era beginning around 2008, and the recent 2022–2024 price surge attributed to the Cocoa Swollen Shoot Virus Disease (CSSVD)

outbreak and other supply-side disruptions in West Africa (Institute 2024; Organization 2024). The sample also includes approximately one year of high-frequency data (February 3, 2025, through January 12, 2026, comprising 20,791 five-minute price observations), which enables validation of the Yang-Zhang volatility estimator against realized volatility computed from intraday returns.

Table 10 in Appendix A summarizes the coverage and key characteristics of the cocoa futures data.

For each trading day t , the dataset provides five price observations: the opening price O_t , the highest price during the trading session H_t , the lowest price L_t and the closing price C_t . Log returns $r_t = \ln(C_t/C_{t-1})$ are computed for descriptive statistics and as inputs to the GARCH benchmark.

From the daily OHLC prices, we construct the Yang-Zhang volatility series (Equation (1)), which serves as the dependent variable in all forecasting models and as the realized measure against which forecasts are evaluated.

For the one-year period with available intraday data, we compute realized volatility as the sum of squared five-minute log returns:

$$RV_t = \sum_{i=1}^M r_{t,i}^2, \quad (20)$$

where $r_{t,i}$ is the log return for the i -th five-minute interval on day t , and M is the number of five-minute intervals in the trading day (approximately 78 intervals for a 6.5-hour trading session). This realized volatility measure serves as the benchmark in a Mincer-Zarnowitz validation of the Yang-Zhang estimator over the recent subsample.

On data quality, the daily series contain no missing values for trading days; non-trading days are excluded. Overnight returns spanning weekends or holidays are not rescaled by calendar days elapsed, as the larger return reflects genuine accumulated risk rather than a measurement artefact.

4.2 ENSO Climate Data

As discussed in Section 2.5, ENSO is hypothesised to provide a physically motivated predictor for cocoa volatility at long horizons. We measure ENSO activity using the Niño 3.4 index, a measure of sea surface temperature anomalies in the central equatorial Pacific (5°N–5°S, 170°W–120°W), computed as a three-month running mean relative to a climatological base period. Monthly values are obtained from the NOAA Climate Prediction Center, calculated using the ERSSTv6 dataset and publicly available at <https://psl.noaa.gov/data/correlation/nina34.data>.

Our sample covers January 1985 through December 2025, providing 492 monthly observations. NOAA classifies an El Niño (La Niña) episode when the three-month

running mean of Niño 3.4 anomalies exceeds $+0.5^{\circ}\text{C}$ (falls below -0.5°C) for at least five consecutive overlapping three-month periods (Lindsey 2009). In the regression models (Equation 7), ENSO_t enters as the continuous Niño 3.4 value rather than a categorical regime indicator, preserving the full variation in anomaly magnitude.

The sample contains 11 El Niño and 12 La Niña episodes over the 1985–2025 period. Figure 3 plots the Niño 3.4 index over the cocoa sample period; red shading marks El Niño episodes (index $> +0.5^{\circ}\text{C}$) and blue shading marks La Niña episodes (index $< -0.5^{\circ}\text{C}$), with the horizontal dashed lines indicating the NOAA classification thresholds.

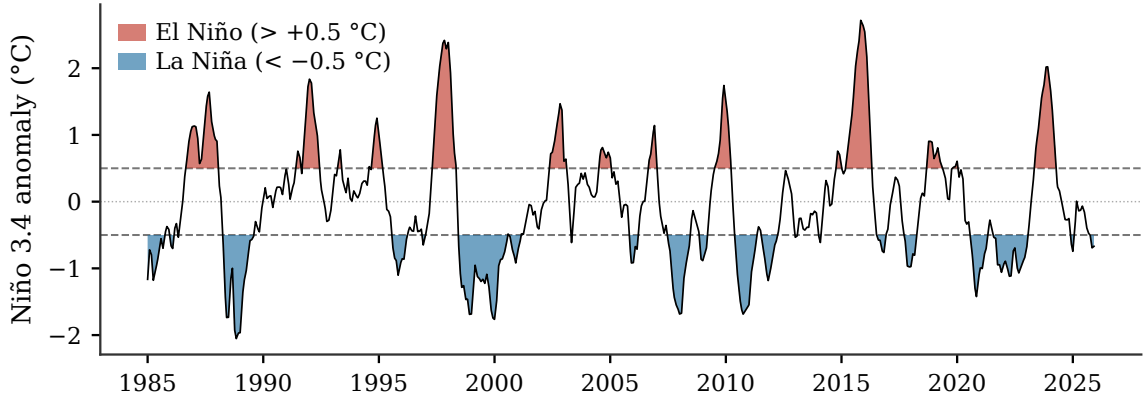


Figure 3: Niño 3.4 Sea Surface Temperature Anomaly (1985–2025)

Monthly Niño 3.4 index (three-month running mean, ERSSTv6 base period). Red: El Niño episodes (anomaly $> +0.5^{\circ}\text{C}$ for at least five consecutive months); blue: La Niña (anomaly $< -0.5^{\circ}\text{C}$).

Dashed lines: $\pm 0.5^{\circ}\text{C}$ thresholds. Source: NOAA Climate Prediction Center.

Since the index is monthly while cocoa futures trade daily, each trading day inherits the Niño 3.4 value of its calendar month, as described in Section 3.2. The data contain no missing values through December 2025, covering the full cocoa futures sample period.

4.3 Descriptive Statistics and Stylized Facts

This subsection presents the empirical characteristics of cocoa returns and volatility, establishing the stylized facts that motivate the modeling choices in Section 3.

Figure 4 displays the daily closing price of the front-month cocoa futures contract over the full sample period. The series exhibits several distinct episodes. From the mid-1980s through the early 2000s, prices declined from their earlier highs and stabilized, fluctuating around a mean of approximately 1,300 GBP per tonne through the 1990s and 2000s. The 2010s showed gradual price increases, followed by the recent 2022–2024 surge reaching over 10,000 GBP per tonne, unprecedented in the modern trading history of cocoa, driven by the CSSVD outbreak in key producing regions.

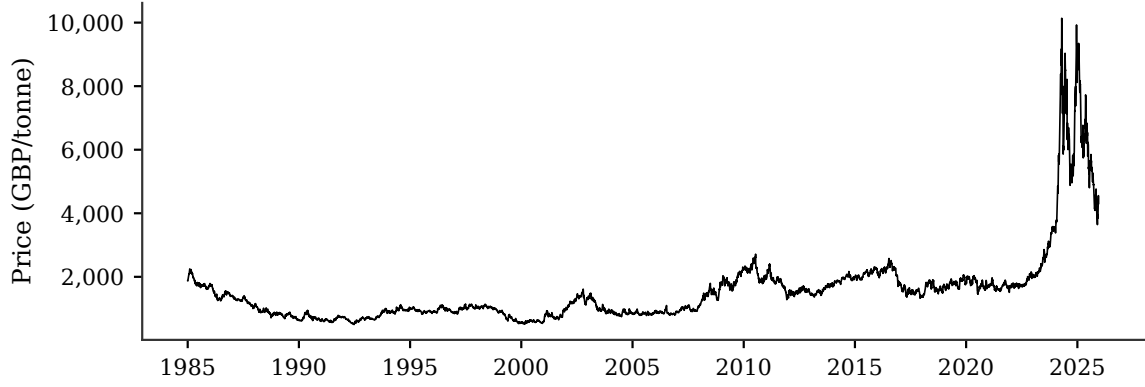


Figure 4: ICE London Cocoa Futures Daily Closing Price (1985–2025)

Front-month continuous contract (LCCc1), daily closing prices in GBP per metric tonne. Source: Refinitiv Workspace. Sample: 2 January 1985 – 15 December 2025 (10,376 trading days).

Table 2 presents descriptive statistics for daily log-returns. The mean daily return is slightly positive (0.008%), corresponding to an approximate annualized mean return of 2% over the full sample. However, returns exhibit substantial volatility: the standard deviation of 1.82% implies an annualized volatility of approximately 29%, consistent with cocoa’s classification as a high-volatility commodity. The distribution of returns deviates sharply from normality. Negative skewness (-0.37) indicates a left tail longer than the right, reflecting the tendency for large negative price movements to occur more frequently than large positive ones. Excess kurtosis of 9.06, formally confirmed by the Jarque-Bera statistic of 35,652 ($p < 0.001$), documents fat tails: extreme returns occur far more often than a Gaussian distribution predicts, implying that volatility models assuming Gaussian innovations may understate tail risk.

Turning to the volatility series, Figure 5 displays the Yang-Zhang volatility estimator (annualized, in percent) computed from daily OHLC data. The figure reveals pronounced volatility clustering: periods of high volatility persist for extended intervals before reverting to lower levels. The 2022–2024 supply crisis stands out as the dominant volatility regime in the sample, with annualized volatility exceeding 100% at its peak. Between this episode and the mid-1980s, volatility fluctuates around a lower baseline with occasional spikes corresponding to supply shocks, weather events, and financial market turbulence (e.g., the 2008 financial crisis).

The persistence of volatility justifies the HAR model’s multi-horizon structure: current volatility depends not only on yesterday’s volatility (daily component) but also on average volatility over the past week and month (weekly and monthly components). The clustering pattern also motivates quantile regression: during high-volatility regimes, the tails of the volatility distribution shift outward, and point forecasts of the mean may underestimate the risk of extreme outcomes.

The regression models in Section 3 require stationary inputs. Augmented Dickey–

Table 2: Descriptive Statistics for Daily Log>Returns

Statistic	Value
Observations	10,375
Mean (%)	0.008
Median (%)	0.000
Std. Deviation (%)	1.816
Minimum (%)	−25.385
Maximum (%)	11.585
25th Percentile (%)	−0.903
75th Percentile (%)	0.915
Skewness	−0.368
Excess Kurtosis	9.057
Jarque-Bera	35,652
<i>p</i> -value	< 0.001

Daily log-returns $r_t = \ln(C_t/C_{t-1})$, ICE London cocoa front-month contract (LCCc1), January 1985–December 2025. Values in percentage points. Excess kurtosis reported (normal distribution: 0). Jarque-Bera tests joint null of zero skewness and zero excess kurtosis.

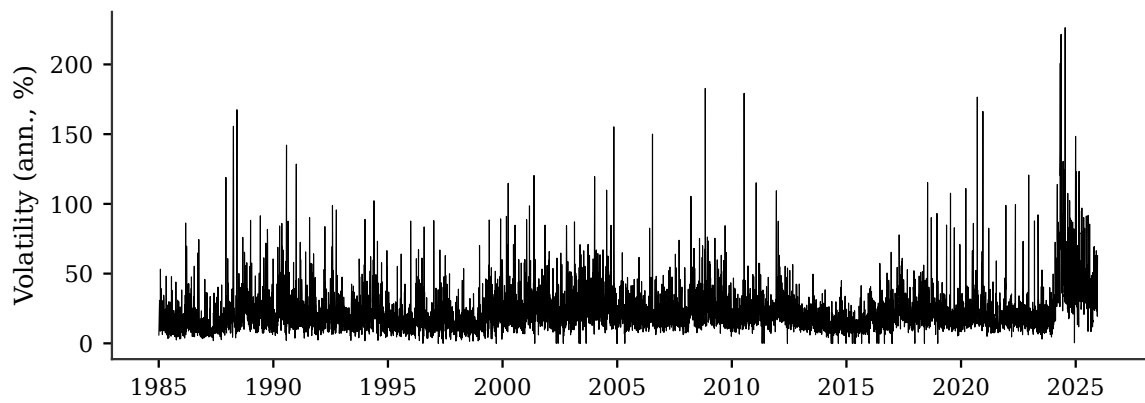


Figure 5: Yang-Zhang Annualized Volatility Time Series (1985–2025)

Yang-Zhang volatility (Yang and Zhang 2000) computed from daily OHLC prices, annualized (percent). Combines overnight, intraday, and close-to-close variance components.

Fuller and Phillips–Perron tests reject the unit root null at the 1% level for all three series (Table 11, Appendix A), justifying the use of standard OLS and quantile regression without differencing or cointegration corrections.

The data described in this section provide the foundation for the empirical analysis. The extended daily sample (1985–2025) ensures sufficient variation in both volatility regimes and ENSO phases, while the intraday subsample enables proxy validation. A Mincer-Zarnowitz regression of intraday realized volatility on the Yang-Zhang estimator (Table 12, Appendix A) yields $R^2 = 0.847$ and $\hat{\beta} = 0.836$ at the 22-day frequency relevant to the HAR monthly component, confirming strong co-movement; the lower $\hat{\beta}$ at the daily frequency reflects the overnight variance component that intraday RV does not capture. The next section presents the empirical results.

5 Empirical Results

This section presents the out-of-sample forecasting results, organized around the two research questions. Section 5.1 compares mean forecast accuracy across the three baseline models. Section 5.2 evaluates distributional forecasts at the upper tail, testing whether HAR-QR outperforms the benchmarks across horizons (RQ1). Section 5.3 assesses whether the Niño 3.4 climate index adds forecast value at long horizons (RQ2). Section 5.4 tests the statistical significance of the observed differences using Model Confidence Sets. Section 5.5 subjects the ENSO result to three diagnostics addressing temporal stability, lag structure, and cross-commodity replication. Section 5.6 synthesizes the findings. All results use a rolling estimation window of 2,520 trading days with daily re-estimation. All models are evaluated on a common sample at each horizon, yielding between 7,306 (twelve months) and 7,737 (one day) out-of-sample forecast–realization pairs.

5.1 Mean Forecast Comparison

Table 3 reports the QLIKE loss for the three mean forecast models across all six horizons. The ranking is unambiguous: HAR-OLS achieves the lowest (best) QLIKE at every horizon, followed by GARCH(1,1), with historical volatility uniformly last.

Table 3: Mean forecast QLIKE loss by model and horizon

Model	1d	1w	1m	3m	6m	12m
Historical Vol	−7.156	−7.362	−7.399*	−7.353	−7.313	−7.266
GARCH(1,1)	−7.262	−7.383	−7.402*	−7.374	−7.341	−7.323
HAR-OLS	−7.291	−7.458	−7.486*	−7.456	−7.405	−7.360

QLIKE loss (Patton 2011); more negative indicates better performance. Bold: best model at each horizon. *Best horizon for each model.

The gap between HAR-OLS and GARCH is largest at the one-month horizon (0.084 QLIKE units) and narrowest at one day (0.029 units). The monthly HAR component ($\sigma_t^{(m)}$) maintains a correlation of 0.66 with the twelve-month volatility target, compared to 0.37 for the daily component, an empirical fact that points to the source of HAR-OLS’s long-horizon advantage.

GARCH(1,1) outperforms historical volatility at every horizon. The margin is widest at one day (0.106 QLIKE units), where GARCH’s daily updating reacts to volatility shocks that a trailing 252-day average smooths away, nearly vanishes at one month (0.003 units), and widens again toward twelve months (0.057 units). All three models share a common horizon pattern: loss improves from the one-day to the one-month horizon, reflecting the smoothing effect of averaging in the target $\bar{\sigma}_{t:t+h}$, before degrading at three through twelve months as the autoregressive signal weakens. Even

HAR-OLS degrades beyond one month, which motivates the search for additional predictive signals in Section 5.3. The mechanism behind the GARCH and HAR ranking is interpreted in Section 6.1.

The QLIKE ranking is broadly confirmed by MSE and MAE. Figure 6 plots all three loss functions across horizons. The HAR family achieves the lowest MSE and MAE at every horizon, and from one week onward GARCH(1,1) records the highest error under both metrics, a direct consequence of mean-reversion in its multi-step point forecasts. Under QLIKE, by contrast, GARCH ranks second throughout and stays ahead of historical volatility even at twelve months. The HAR family’s lead over both benchmarks is preserved regardless of how point forecast errors are penalized.

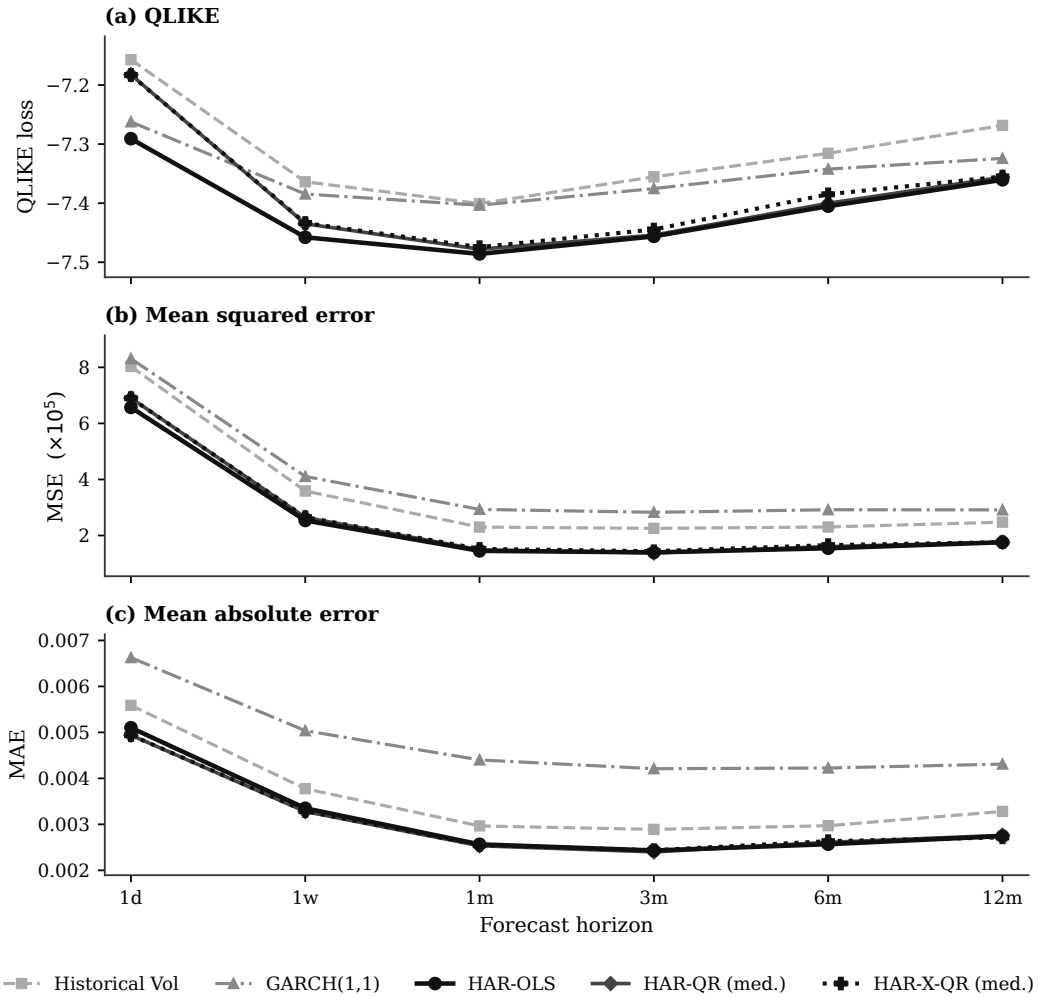


Figure 6: Robustness check: point forecast accuracy under three loss functions. Each panel plots mean (or median) forecast loss across horizons; in all three panels a lower line is better. QLIKE is shown on its native scale, where a more negative value is a smaller loss (Patton 2011). MSE scaled by 10^5 for readability. The two HAR-QR medians are included to show that the full HAR family’s point-forecast behaviour coincides with HAR-OLS. The HAR family separates from both benchmarks under MSE and MAE at all horizons; under QLIKE, GARCH(1,1) ranks second behind HAR-OLS.

5.2 Quantile Forecast Comparison (RQ1)

Mean forecasts summarize only the center of the conditional distribution. For risk assessment, the upper tail matters: a model that accurately forecasts average volatility may still understate the probability of extreme outcomes. We now compare five models on their ability to forecast the 95th percentile of future volatility. Historical volatility uses the empirical quantile of the trailing 252-day distribution of h -day average volatility. GARCH(1,1) quantiles follow the closed-form Gaussian approach (Equation 13). HAR-OLS produces pseudo-quantile forecasts from its in-sample residual distribution (Equation 14). HAR-QR estimates the quantile directly through quantile regression on the HAR structure, and HAR-X-QR augments this with the Niño 3.4 index.

Table 4 reports the quantile loss at $\tau = 0.95$ for all five models.

Table 4: Upper-tail quantile loss ($\tau = 0.95$) by model and horizon

Model	1d	1w	1m	3m	6m	12m
Historical Vol	0.001281	0.000830	0.000691*	0.000809	0.001033	0.001462
GARCH(1,1)	0.001283	0.000828	0.000609*	0.000629	0.000668	0.000736
HAR-OLS	0.001229	0.000746	0.000536*	0.000558	0.000608	0.000786
HAR-QR	0.001159	0.000682	0.000500*	0.000557	0.000704	0.001041
HAR-X-QR	0.001160	0.000685	0.000521*	0.000580	0.000711	0.001007

Quantile loss (Equation (18)). Lower values indicate better forecasts. Bold: best model at each horizon. *Best horizon for each model.

Four patterns emerge from this comparison.

First, direct quantile estimation wins at short and medium horizons. HAR-QR achieves the lowest quantile loss at one day, one week, one month, and (marginally) three months. The advantage over the benchmarks is not trivial: at one month, the best benchmark (GARCH, 0.000609) is 22% above HAR-QR (0.000500), and at one day historical volatility (0.001281) is 11% above HAR-QR (0.001159). Against the HAR-OLS pseudo-quantile, HAR-QR improves by 6% at one day and 7% at one month; at three months the two are essentially tied (0.000557 vs 0.000558).

Second, the ranking reverses at long horizons. At six months the HAR-OLS pseudo-quantile takes the lead at 0.000608, which is 14% below HAR-QR. At twelve months GARCH(1,1) becomes the best model outright, with a quantile loss (0.000736) that is 29% below HAR-QR and 27% below HAR-X-QR. The advantage of estimating the quantile directly on rolling windows is thus a short-horizon phenomenon; at the longest horizons, simpler quantile constructions anchored to the unconditional distribution prove more robust. Whether these long-horizon reversals are statistically meaningful is assessed in Section 5.4.

Third, historical volatility collapses at long horizons. Its empirical trailing quantile is competitive at one month (0.000691) but deteriorates rapidly beyond that, reaching

0.001462 at twelve months — 40% above HAR-QR and 99% above GARCH. A trailing one-year window cannot anticipate distributional shifts over a further one-year forecast window, so violations cluster exactly where the upper tail matters most.

Fourth, ENSO augmentation does not separate HAR-X-QR from HAR-QR in this model’s favor except at twelve months. The two models are virtually indistinguishable at one day and one week (differences $< 0.5\%$); ENSO *worsens* the upper-tail forecast at one month (-4.2%), three months (-4.2%), and six months (-1.0%); and only at twelve months does a modest improvement of 3.3% appear. This horizon dependence is analyzed in detail in Section 5.3.

Table 4 isolates the upper tail, where the long-horizon ranking reverses in favour of the benchmarks. The full five-model comparison at the median and $\tau = 0.75$, reported in Appendix A (Table 13), shows that this reversal is confined to the upper tail: at $\tau = 0.50$ and $\tau = 0.75$ the HAR family (HAR-QR or HAR-OLS) records the lowest quantile loss at every horizon, and neither benchmark is ever competitive. The twelve-month advantage of GARCH is thus a tail phenomenon, not a general long-horizon one.

Figure 7 illustrates the distributional forecasts of the HAR-X-QR model, displaying the 50–75% and 75–95% quantile bands alongside the median forecast ($\tau = 0.50$) and Yang-Zhang volatility. Panels (a1) and (b1) cover 1995–2022; panels (a2) and (b2) show the 2023–2025 crisis window on its own scale. The figure complements the loss-function evidence by showing how the model’s forecast distribution moves through regimes.

At the one-month horizon (panels a1, a2), the quantile bands widen and narrow in step with Yang-Zhang volatility, demonstrating the HAR structure’s ability to adapt to changing variance regimes; during the 2024 crisis peak the bands follow the realized spike with a short lag. At the twelve-month horizon (panels b1, b2), the bands are smoother, reflecting the averaging inherent in long-run projection. The crisis panel (b2) exposes the central weakness of the long-horizon forecast: Yang-Zhang volatility rises above the $\tau = 0.95$ boundary for an extended stretch of 2023–24 before the forecast distribution adjusts, and the bands widen only after the spike has materialized. This delayed adjustment is reflected in the violation rates reported in Table 5.

Table 5 reports violation rates for the upper quantile ($\tau = 0.95$) by horizon for both HAR-QR and HAR-X-QR. Under correct unconditional coverage, the violation rate should equal $1 - \tau = 5\%$. HAR-QR achieves near-target coverage at one day and one week (5.17% and 5.39%), then deteriorates monotonically as the horizon extends: 6.35% at one month, 8.15% at three months, 11.45% at six months, and 14.63% at twelve months. ENSO augmentation leaves short-horizon coverage essentially unchanged but worsens calibration at every horizon from one month onward: HAR-X-QR violation rates reach 14.15% at six months and 15.89% at twelve months. Formal

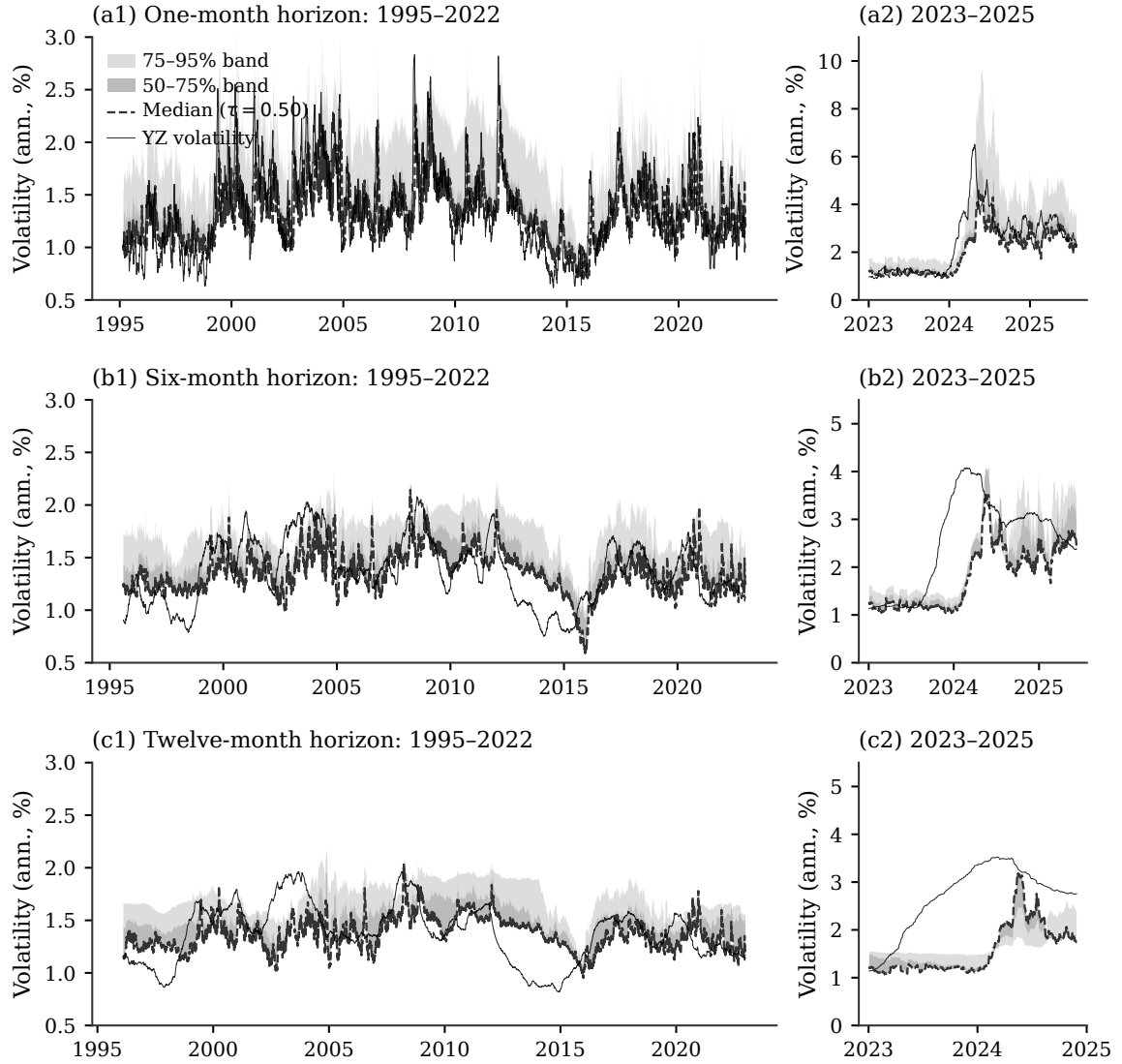


Figure 7: HAR-X-QR distributional forecasts: quantile bands and Yang-Zhang volatility

HAR-X-QR quantile forecasts at $\tau \in \{0.50, 0.75, 0.95\}$ plotted against Yang-Zhang volatility (solid black). Dark gray band: 50–75% interval; light gray band: 75–95% interval; dashed line: median ($\tau = 0.50$). Rows: one-month (a) and twelve-month (b) horizons. Left column: 1995–2022; right column: 2023–2025 (separate scale).

Christoffersen tests on non-overlapping subsequences are reported in Table 15 (Appendix A); the calibration–discrimination tradeoff behind these numbers is taken up in Section 6.4. The following subsection examines whether augmenting HAR-QR with the ENSO climate index improves long-horizon upper-tail accuracy.

Table 5: Upper-quantile ($\tau = 0.95$) coverage by horizon: HAR-QR and HAR-X-QR

Horizon	N	HAR-QR rate (%)	HAR-X-QR rate (%)
1d	7,737	5.17	5.14
1w	7,733	5.39	5.41
1m	7,716	6.35	7.14
3m	7,672	8.15	9.68
6m	7,570	11.45	14.15
12m	7,306	14.63	15.89

A violation occurs when Yang-Zhang volatility exceeds the $\tau = 0.95$ quantile forecast; the nominal target is 5% ($= 1 - \tau$). N : forecast–realization pairs. Both models are evaluated on the identical sample at each horizon, restricted to dates with an available Niño 3.4 reading (the ENSO series ends in July 2025, five months before the daily cocoa sample).

5.3 ENSO Augmentation: HAR-X-QR (RQ2)

Table 6 reports the percentage change in quantile loss when moving from HAR-QR to HAR-X-QR across all three quantile levels and six horizons. Positive values indicate improvement (lower loss under HAR-X-QR).

Table 6: Quantile loss improvement (%) from ENSO augmentation (HAR-X-QR vs. HAR-QR)

τ	1d	1w	1m	3m	6m	12m
0.50	+0.00	−0.32	−0.62	−1.54	−0.99	+1.62
0.75	−0.02	−0.31	−1.34	−3.34	−3.53	−6.72
0.95	−0.09	−0.43	−4.20	−4.24	−1.03	+3.29

Improvement (%) = $(1 - \text{QL}_{\text{HAR-X}}/\text{QL}_{\text{HAR}}) \times 100$; positive = lower loss for HAR-X-QR.

The dominant feature of this table is the prevalence of negative entries: in fifteen of the eighteen quantile–horizon combinations, adding the Niño 3.4 index *worsens* the out-of-sample forecast. At short horizons (one day to one week) the effect is negligible in either direction — the additional coefficient neither helps nor hurts materially. From one month onward, the extra parameter degrades performance at almost every quantile level, with the deterioration most pronounced at $\tau = 0.75$, where it grows monotonically with the horizon and reaches −6.7% at twelve months.

The only improvements appear at the twelve-month horizon: +3.3% at the upper tail ($\tau = 0.95$) and +1.6% at the median. This single favourable cell is the entire

empirical case for RQ2, and three properties place it under suspicion before any further testing. It is modest in magnitude; it is isolated, surrounded by deterioration at every shorter horizon; and it earns HAR-X-QR no top rank in the Model Confidence Set at any horizon (Section 5.4). It must also be read against the benchmark comparison in Table 4: even with the ENSO gain, HAR-X-QR (0.001007) remains 37% above GARCH(1,1) (0.000736) at the twelve-month upper tail. The climate index narrows the gap to the best model; it does not close it. The gain cannot be dismissed on magnitude alone, however, because it falls precisely at the twelve-month procurement horizon that motivates the thesis. It therefore warrants a direct test of whether it reflects a stable predictive relationship or a sample-specific regime effect — the subject of Section 5.5.

5.4 Statistical Significance: Model Confidence Sets

The preceding sections compare models by point estimates of loss functions. To assess whether the observed differences are statistically significant, we apply the Model Confidence Set (MCS) procedure of Hansen et al. (2011). The MCS identifies the set $\mathcal{M}^*$ of models whose forecasting ability cannot be distinguished from that of the best model at a given confidence level. We use the MCS procedure and configuration described in Section 3.4.

Table 7 reports MCS results for QLIKE loss. HAR-OLS is the sole member of $\mathcal{M}^*$ at the one-day, one-week, and one-month horizons ($p = 1.00$), confirming that its mean forecast superiority is statistically significant at short and medium horizons. From three months onward, all five models enter $\mathcal{M}^*$: at long horizons the loss differentials, while still favoring HAR-OLS in point terms, are no longer large relative to their sampling variation. GARCH(1,1) and historical volatility are thus excluded through one month and statistically indistinguishable from the best model thereafter.

Table 8 presents MCS results for quantile loss at $\tau = 0.95$, the central evaluation criterion for distributional forecasting. At the one-day and one-week horizons, HAR-QR holds the top rank ($p = 1.00$) and $\mathcal{M}^*$ contains only the two direct quantile regression models: historical volatility, GARCH, and the HAR-OLS pseudo-quantile are all excluded, confirming that explicit quantile estimation provides a statistically significant advantage where volatility dynamics are most state-dependent. From one month onward the sample no longer discriminates sharply: at one month all five models enter $\mathcal{M}^*$, and from three months onward only historical volatility is excluded.

The migration of the top rank across horizons mirrors the point estimates. HAR-QR is ranked first through three months; at six months HAR-OLS takes $p = 1.00$; and at twelve months GARCH(1,1) is the top-ranked model. Because all of these models (and both QR variants) remain in $\mathcal{M}^*$ at long horizons, the data do not permit a

Table 7: Model Confidence Set: QLIKE loss (mean forecasts)

Model		Horizon					
		1d	1w	1m	3m	6m	12m
Historical Vol	p	0.034	0.096	0.097	0.132	0.451	0.218
	$\mathcal{M}^*$				✓	✓	✓
GARCH(1,1)	p	0.034	0.096	0.030	0.129	0.451	0.460
	$\mathcal{M}^*$				✓	✓	✓
HAR-OLS	p	1.000	1.000	1.000	1.000	1.000	1.000
	$\mathcal{M}^*$	✓	✓	✓	✓	✓	✓
HAR-QR (med.)	p	0.019	0.096	0.097	0.514	0.451	0.715
	$\mathcal{M}^*$				✓	✓	✓
HAR-X-QR (med.)	p	0.019	0.096	0.097	0.132	0.451	0.715
	$\mathcal{M}^*$				✓	✓	✓

MCS p -values (Hansen et al. 2011): max- t statistic, 10,000 bootstrap replications, 90% confidence.

✓: $\mathcal{M}^*$ membership. Bold: $p = 1.00$ (best-ranked model). HAR-QR and HAR-X-QR enter as median ($\tau = 0.50$) forecasts.

statistically significant ranking among them; but the consistent shift of the best point performance away from direct quantile regression at six and twelve months marks the QR advantage as a short-horizon result. At no horizon does HAR-X-QR hold the top rank, so the MCS provides no statistical support for the ENSO augmentation either.

Table 8: Model Confidence Set: quantile loss at $\tau = 0.95$

Model		Horizon					
		1d	1w	1m	3m	6m	12m
Historical Vol	p	0.066	0.053	0.105	0.091	0.041	0.012
	$\mathcal{M}^*$			✓			
GARCH(1,1)	p	0.001	0.001	0.105	0.159	0.591	1.000
	$\mathcal{M}^*$			✓	✓	✓	✓
HAR-OLS	p	0.066	0.071	0.678	0.950	1.000	0.610
	$\mathcal{M}^*$			✓	✓	✓	✓
HAR-QR	p	1.000	1.000	1.000	1.000	0.591	0.216
	$\mathcal{M}^*$	✓	✓	✓	✓	✓	✓
HAR-X-QR	p	0.279	0.386	0.678	0.718	0.548	0.216
	$\mathcal{M}^*$	✓	✓	✓	✓	✓	✓

Same MCS configuration as Table 7. ✓: $\mathcal{M}^*$ membership at 90% confidence. Bold: $p = 1.00$ (best-ranked model).

5.5 Robustness of the ENSO Signal

The preceding results leave RQ2 resting on a single number: the 3.3% upper-tail improvement at the twelve-month horizon. The decisive question is whether this gain accumulates steadily across the three decades of ENSO cycles in the sample — the signature of a structural climate channel — or materialises within a single episode. The first diagnostic below answers this directly, and without any researcher-chosen break date; it is therefore the primary test, and the same construction that allows it to dispense with date choices is what makes its verdict robust to the objectivation concerns that motivate the design. The two diagnostics that follow — lag structure and cross-commodity replication — address distinct generalisability concerns raised at the design stage and corroborate the verdict from independent angles. Together the three test three ways in which the full-sample gain could be sample-specific rather than generalisable: temporal concentration in one episode, a lag structure inconsistent with the proposed mechanism, and failure to replicate across comparable commodities.

Temporal stability: expanding-window diagnostic

To assess whether the ENSO gain is stable through time or instead driven by a particular episode in the sample, we trace the *cumulative* gain as the evaluation horizon advances through time. For each calendar month T in the out-of-sample period, define

$$\Delta\text{QL}(T) = \text{QL}_{\text{HAR-X-QR}}(\{t \leq T\}) - \text{QL}_{\text{HAR-QR}}(\{t \leq T\}),$$

where each loss is computed over all forecast–realisation pairs with evaluation date up to and including T . Negative values indicate that ENSO augmentation has been beneficial on a cumulative basis up to that point. The construction is fully recursive: no break date is assumed or imposed, and the curve spans the complete out-of-sample window without any ex-post partitioning.

Reading the curves requires keeping their construction in mind. Each is a difference of two running means — cumulative HAR-X-QR loss minus cumulative HAR-QR loss, both averaged over every forecast–realisation pair up to T — so it is the *slope*, not the level, that carries the sign of ENSO’s contribution in each subperiod. A downward-sloping stretch is one in which augmentation reduces loss relative to baseline; a flat stretch is one in which it has no net effect; and a near-vertical step marks a single episode in which the two models diverge sharply. Two construction artifacts must be read through. First, a running mean is most sensitive to individual observations early in the sample, when few pairs have accumulated, so the wide swings before roughly 2005 reflect the small evaluation sample as much as the intensity of the late-1990s ENSO cycle. Second, once an episode has displaced the cumulative mean, the curve

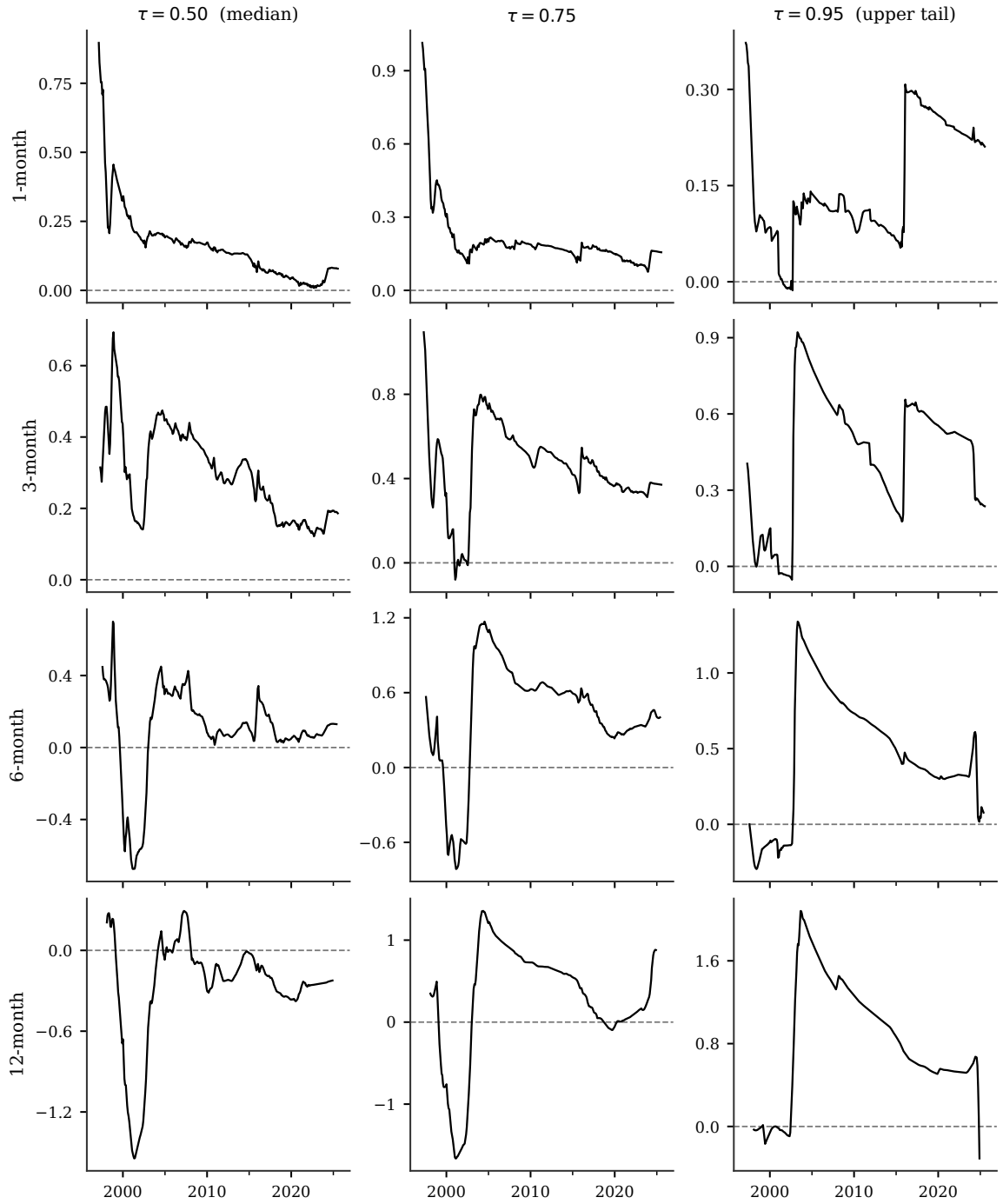


Figure 8: Expanding-window cumulative ENSO augmentation gain (ΔQL) across horizons and quantile levels

Each panel shows $\Delta QL(T)$ over all out-of-sample pairs with evaluation date $\leq T$; negative values indicate cumulative net gain from ENSO augmentation. Rows: horizons (1, 3, 6, 12 months).

Columns: $\tau \in \{0.50, 0.75, 0.95\}$.

drifts back toward zero over the following years even when ENSO contributes nothing further: that post-episode slope is dilution by later near-neutral observations, not a reversal of sign. A structural channel must therefore appear as a sustained downward drift across many years — not as a step followed by passive decay.

The figure is laid out as a matrix — horizons down the rows, quantile levels across the columns — precisely so that the absence of a structural signal can be read off directly. No panel in Figure 8 exhibits the sustained-drift shape a genuine climate channel would produce. In every one of the twelve quantile–horizon combinations the cumulative ΔQL is either flat around zero or dominated by one or two abrupt steps tied to identifiable episodes, with long stretches of no net movement in between. The matrix as a whole therefore delivers the central robustness finding before any single cell is examined: wherever ENSO appears to help, the help arrives in discrete supply-stress episodes rather than accruing across cycles. Two features of the layout sharpen this reading. Moving down any column, the curves grow more step-like as the horizon lengthens, because longer-horizon forecasts overlap a single volatility episode many times and so register it as one large displacement; moving across any row, the $\tau = 0.95$ column shows the sharpest episodic steps, because the upper tail is where a volatility spike registers most forcefully. The one cell that ends below zero — the twelve-month upper tail — is thus also the cell in which a single episode can most easily overturn three decades of accumulated sign.

Read on its own, that cell makes the point concrete. At the twelve-month upper tail — the only quantile–horizon combination with a material full-sample gain — the cumulative ΔQL is *positive* (ENSO augmentation harmful on net) for essentially the entire evaluation period. An adverse episode in the early 2000s pushes the curve sharply upward, after which it declines only gradually as two decades of near-neutral observations dilute the early loss. The single large favourable movement occurs in 2023–24, when realised volatility spiked during the supply crisis while the ENSO state was strongly El Niño; this final episode pulls the cumulative curve just below zero, producing the full-sample endpoint of $\Delta QL = -0.34 \times 10^{-4}$ (the 3.3% gain). The six-month panel tells the same story without the sign flip: its endpoint remains marginally positive ($+0.07 \times 10^{-4}$), because the crisis episode is not quite large enough to offset the accumulated deterioration.

The pre-crisis subsample makes the dependence explicit. Restricting the evaluation to forecasts whose entire h -day target window precedes February 2024 (for the twelve-month horizon, origins up to January 2023), ENSO augmentation *raises* the twelve-month upper-tail quantile loss by 16% ($\Delta QL_{\text{pre}} = +0.52 \times 10^{-4}$, on a pre-crisis loss level of 0.000322). The entire full-sample gain — and more — is contributed by the 2023–24 window.

The other quantile levels reinforce the episodic reading. At the median and $\tau =$

0.75, a favourable episode around the turn of the millennium — plausibly connected to the 1997–98 El Niño, the strongest on record at that time — produces a transient cumulative gain at the longer horizons that dissipates over the following decade. Notably, this same turn-of-the-millennium episode moves the central and upper-tail panels in *opposite* directions: a favourable downward dip at the median and $\tau = 0.75$, but the sharp upward step that dominates the twelve-month upper-tail panel. The same ENSO alignment that improved the central forecast worsened the upper-tail forecast — a sign reversal across quantiles consistent with the discrimination–calibration tension analysed in Section 6.4. At $\tau = 0.75$ the curve then turns and remains adverse, ending at $+0.88 \times 10^{-4}$ at twelve months (the -6.7% entry in Table 6); at the median the endpoint is indistinguishable from zero. At no quantile–horizon combination does the diagnostic show a steadily accumulating gain across successive ENSO cycles: every favourable movement is confined to a specific supply-stress episode, and the dominant one is the 2022–2024 crisis.

Two corroborating checks: lag structure and cross-commodity replication

Two further checks, requested at the design stage when the apparent gain was several times larger, confirm the temporal verdict from independent angles; both are now reported compactly, with full tables in Appendix A. *Lag structure*: if ENSO acted through the agronomic-transmission chain (sea-surface anomaly $\rightarrow$ West African rainfall $\rightarrow$ harvest shortfall $\rightarrow$ supply tightening $\rightarrow$ price volatility), predictive power should strengthen as the lag approaches the 6–12 months over which that chain unfolds. The opposite holds (Table 16): where ENSO helps at the twelve-month upper tail it does so only contemporaneously (lag 0) or at lag 1, while the agronomically motivated lags of three and six months are *worse* than the HAR-QR baseline with no ENSO term. *Cross-commodity replication*: a genuine Pacific climate channel should lift forecasts for other tropical soft commodities with the same exposure. It does not (Table 17): on the identical HAR-X-QR specification, ENSO *worsens* twelve-month upper-tail forecasts for Arabica coffee and produces only an economically negligible gain for raw sugar (1.6%). The one commodity that shows any gain is the one whose sample contains an idiosyncratic supply crisis.

Summary assessment of RQ2 robustness

The diagnostics yield a consistent verdict, and the primary one settles it: the full-sample twelve-month upper-tail gain of 3.3% is the residual of a cumulative deterioration spanning most of the evaluation period and a single favourable episode in 2023–24, and in the pre-crisis subsample ENSO augmentation makes the same forecast 16% worse. The two corroborating checks point the same way — predictive power, where

present, sits at contemporaneous rather than agronomic lags, and the result does not reproduce on coffee or sugar. The interpretation of this combined pattern, and what it implies for the answer to RQ2, is taken up in Section 6.2.

5.6 Summary of Findings

Table 9 consolidates the key metrics for all models across the horizon spectrum. Three conclusions emerge.

Table 9: Consolidated model comparison across horizons

Metric	Model	Horizon				
		1d	1m	3m	6m	12m
QLIKE	Historical Vol	−7.156	−7.399*	−7.353	−7.313	−7.266
	GARCH(1,1)	−7.262	−7.402*	−7.374	−7.341	−7.323
	HAR-OLS	−7.291	−7.486*	−7.456	−7.405	−7.360
QL ($\tau=0.95$)	Historical Vol	0.001281	0.000691*	0.000809	0.001033	0.001462
	GARCH(1,1)	0.001283	0.000609*	0.000629	0.000668	0.000736
	HAR-OLS	0.001229	0.000536*	0.000558	0.000608	0.000786
	HAR-QR	0.001159	0.000500*	0.000557	0.000704	0.001041
	HAR-X-QR	0.001160	0.000521*	0.000580	0.000711	0.001007

Bold: best model at each horizon. *Best horizon for each model. QLIKE: more negative is better; QL: lower is better. 1w horizon omitted for compactness.

First, RQ1 receives a horizon-dependent answer. At short-to-medium horizons the evidence for direct quantile regression is strong: HAR-QR achieves the lowest upper-tail quantile loss from one day through three months, and at one day and one week it is the top-ranked model in a Model Confidence Set from which all three benchmarks are excluded (Table 8). At long horizons the advantage disappears: HAR-OLS overtakes HAR-QR at six months and GARCH(1,1) is the best upper-tail forecaster at twelve months, 29% below HAR-QR in quantile loss. The long-horizon differences are not statistically significant — all models except historical volatility share $\mathcal{M}^*$ from three months onward — but the migration of the top MCS rank from HAR-QR (one day to three months) to HAR-OLS (six months) to GARCH (twelve months) consistently locates the value of explicit quantile estimation at the short end. For mean forecasts, HAR-OLS dominates at every horizon and the HAR family separates cleanly from GARCH under MSE and MAE (Figure 6).

Second, RQ2 is answered in the negative. ENSO augmentation worsens the upper-tail forecast at every horizon from one month to six months and delivers only a modest 3.3% improvement at twelve months (Table 6). The robustness diagnostics in Section 5.5 show that this residual gain is concentrated entirely in the 2023–2024 crisis window (pre-crisis, ENSO makes the twelve-month forecast 16% worse), follows a lag

structure inconsistent with the agronomic-transmission hypothesis, and does not reproduce on comparable tropical commodities. The interpretation of this combined pattern is developed in Section 6.2.

Third, coverage analysis reveals a systematic challenge at long horizons. Both direct quantile regression models produce violation rates far above the nominal 5% at six months (11.45% and 14.15%) and twelve months (14.63% and 15.89%), with the deterioration concentrated in the crisis episode, during which realized volatility remained above the $\tau = 0.95$ boundary for an extended period (Figure 7, panel b2). ENSO augmentation worsens calibration at every horizon from one month onward. The implications of this calibration pattern are discussed in Section 6.4.

6 Discussion

The empirical results in Section 5 establish two findings that require interpretation. Direct quantile regression on the HAR structure delivers a statistically significant upper-tail advantage at short horizons but is overtaken by simpler quantile constructions — the HAR-OLS pseudo-quantile at six months and the GARCH(1,1) Gaussian quantile at twelve months — as the horizon extends (RQ1), and the Niño 3.4 climate index provides no robust forecast improvement at any horizon, its modest twelve-month full-sample gain being concentrated entirely in the 2022–2024 supply crisis (RQ2). This section explains why these patterns arise, how they relate to the existing literature on volatility forecasting and ENSO–commodity linkages, and what they imply for the calibration of distributional forecasts, the procurement perspective that motivated the choice of cocoa, and the directions for further research.

6.1 Why the HAR Family Dominates

The dominance of the HAR family in mean forecasting is structural rather than incidental. Corsi (2009) shows that standard short-memory volatility models, including GARCH, fail to reproduce the scaling behaviour of volatility under temporal aggregation: their exponential memory decay erases shocks too rapidly, and multi-step forecasts converge toward the unconditional variance within a few weeks. The HAR structure approximates the hyperbolic decay that empirical volatility exhibits by stacking heterogeneous lags at daily, weekly, and monthly frequencies, without imposing a fractional integration parameter. This matters with particular force for cocoa, which Alfeus and Nikitopoulos (2022) identify as a long-memory commodity in which monthly volatility carries the most predictive weight. Section 5.1 reports a correlation of 0.66 between the monthly HAR component and the twelve-month volatility target, against 0.37 for the daily component, which is the direct empirical counterpart of Corsi’s scaling argument and explains why HAR-OLS retains its lead even at twelve months while GARCH(1,1) flatlines.

The upper-tail ranking at long horizons inverts this picture, and the inversion is informative about what each model actually estimates. The GARCH quantile forecast (Equation 13) anchors the $\tau = 0.95$ boundary to the dispersion of h -day average volatility over the full estimation window, while its conditional mean is updated daily. At twelve months this construction is conservative but stable: it does not attempt to extrapolate tail dynamics from the recent past alone. The rolling HAR-QR quantile, by contrast, is fitted to the empirical upper tail of a 2,520-day window; when the 2022–2024 crisis pushed realized volatility far outside the support of that window, the fitted quantile surface had no comparable episodes to learn from, and the forecast boundary adjusted only with substantial delay (Figure 7, panel b2). The result is

GARCH's 29% quantile-loss advantage over HAR-QR at the twelve-month upper tail (Table 4): a deliberately wide, slowly-moving quantile loses little in calm decades and avoids the catastrophic under-prediction that dominates the loss of the more adaptive models during the crisis. The advantage is one of point loss only; at twelve months GARCH, HAR-OLS, HAR-QR and HAR-X-QR all belong to the model confidence set (Section 5.4), so the ranking reflects the lowest average loss rather than a statistically separable edge.

Within the HAR family, the comparison between HAR-QR and the HAR-OLS pseudo-quantile addresses a more subtle question: does the conditional upper quantile diverge sufficiently from a shift of the conditional mean to justify the additional estimation cost of quantile regression? Haugom et al. (2016) argue that the gain comes from two sources: direct quantile estimation is robust to distributional misspecification, and the monthly volatility component exerts its strongest influence precisely at the conditional tails. The cocoa results match this mechanism. The HAR-QR advantage over the HAR-OLS pseudo-quantile is largest at the short to medium horizons (9% at one week, 7% at one month, 6% at one day), where volatility dynamics are most state-dependent and the conditional quantile is furthest from the unconditional residual quantile. From six months onward the advantage reverses in point terms: HAR-OLS undercuts HAR-QR by 14% at six months and 24% at twelve months (0.000786 vs. 0.001041), though the two remain statistically indistinguishable in the MCS (Section 5.4). Lyócsa and Stašek (2021) report the same horizon dependence on stock-market data: the value of explicit quantile estimation is concentrated at short horizons where extreme regime states drive the conditional distribution. Coverage at the one-day and one-week horizons is close to the nominal 5% (5.17% and 5.39%). Christoffersen's LR_{uc} test confirms that unconditional coverage is indistinguishable from the nominal 5% at one day ($p = 0.495$), consistent with Haugom's finding for equity markets, now extended to an agricultural commodity. However, the independence component LR_{ind} is rejected at one day ($p = 0.008$), indicating that violations cluster even at the shortest horizon: the conditional coverage test is formally rejected ($p = 0.024$). At longer horizons, overlapping h -day targets make consecutive violation indicators dependent by construction, so the Christoffersen statistics are computed on the non-overlapping subsequence of every h -th forecast (Table 15, Appendix A). On that basis, unconditional coverage is rejected for HAR-QR at six months, while at twelve months the non-overlapping subsequence (28 observations) is too short for an informative test and the raw violation rate of 14.63% is the more telling summary.

6.2 Interpreting the ENSO Signal

The ENSO result requires less interpretive caution than the research design anticipated, because the corrected evidence points uniformly in one direction. Over the full evaluation sample, HAR-X-QR reduces upper-tail quantile loss by 3.3 % at the twelve-month horizon — the only quantile–horizon combination with a gain of any substance — while worsening the forecast at fifteen of the eighteen combinations examined, and it holds the top MCS rank at no horizon. The robustness diagnostics in Section 5.5 show that even the twelve-month gain is fragile: the cumulative gain is negative over the entire pre-crisis period, materialises wholly within the 2022–2024 supply-crisis window, follows a lag structure opposite to what an agronomic transmission chain would predict, and does not replicate on Arabica coffee or raw sugar. The interpretive task is not to explain away a striking effect, but to understand why a mechanism with a plausible physical basis leaves so faint a statistical trace.

The physical channel that motivated the inclusion of ENSO in the first place is plausibly real but is unlikely to be visible at the lag structure assumed by the standard climate-yield literature. Schroth et al. (2016) document that severe El Niño events in the 1980s coincided with drought years that materially affected West African cocoa yields, which suggests a transmission window of six to twelve months from sea-surface anomaly to harvest impact. Our lag-decay diagnostic shows the opposite pattern: predictive power peaks at lag 1 and decays sharply by lag 6. Bergander (2025) provide the missing link: futures prices in their analysis reflect roughly 85 % of the eventual price impact of a drought shock *before* the actual crop loss occurs. If professional traders anticipate ENSO-induced supply disruption and hedge forward accordingly, the volatility footprint of an ENSO anomaly should appear near-contemporaneously with the climate signal, not lagged by the agronomic timescale. The lag-decay pattern is therefore consistent with market anticipation collapsing the physical lag, rather than with the absence of an underlying mechanism.

What anticipation alone cannot explain, however, is the episodic and non-persistent character of the cumulative gain. A stable anticipation channel should deliver a steady forecast improvement across decades of ENSO cycles, not gains that materialise in specific supply-stress episodes and revert in the intervening years. The El Niño episode around the turn of the millennium, which produced a temporary gain at the centre of the distribution that dissipated over the following decade, provides a second illustration of this pattern: even a real ENSO–supply alignment does not create a structural signal that survives into subsequent climate cycles. The plausible bridge between these two observations is the rare-disaster reading proposed by Bonato et al. (2023): ENSO acts as a proxy for the level of uncertainty over future supply, and risk-averse inventory holders respond to high ENSO uncertainty by building stocks, which in turn raises the

convenience yield and amplifies the variance of returns. Under this reading, ENSO is predictive of upper-tail volatility only when the underlying supply system is already close to a stress threshold; in quiet periods, the same Niño 3.4 readings carry no incremental information beyond the HAR lags. The 2022–2024 episode supplies precisely that stress threshold via the CSSVD outbreak and the resulting structural supply shortfall, and the ENSO regressor identifies the El Niño transition that coincides with it. Bouri et al. (2021) find a structurally similar pattern for oil, where the predictive value of ENSO is concentrated at specific horizons (two to four years) and emerges most strongly under longer estimation windows that capture full climate cycles; the horizon over which the signal becomes visible differs across markets, but the regime-conditional character of the effect is shared.

This reading is also consistent with the cross-commodity null. Ubilava (2018) reports that ENSO produces in-sample evidence of predictability for selected non-agricultural commodities but that the out-of-sample relationship typically fails to replicate, and that the effect is most amplified for cocoa during El Niño episodes specifically. The HAR-X-QR results extend this pattern: coffee and sugar share the same equatorial belt and the same Pacific climate exposure, yet the same regressor that produces cocoa’s crisis-concentrated twelve-month gain *worsens* long-horizon coffee forecasts and produces only an economically negligible gain on sugar. A structural climate channel operating uniformly across tropical soft commodities is hard to reconcile with this asymmetry; a regime-state indicator that happens to align with one severe cocoa-specific episode is not. Huang et al. (2022) provide further support for this reading by showing that HAR coefficients differ substantially across market regimes; the ENSO coefficient, on this interpretation, is identified predominantly in the regime where its sign and magnitude happen to align with the realised volatility surprise.

The data therefore support a clear answer to RQ2: the ENSO index does not improve cocoa volatility forecasts in any robust sense. The full-sample twelve-month upper-tail gain of 3.3 % is the residual of a single crisis episode superimposed on decades of mild deterioration, and in the pre-crisis subsample the augmented model is strictly worse. A coherent mechanism, combining market anticipation with a rare-disaster amplification through inventory dynamics, can rationalise both why whatever predictive content ENSO carries appears near-contemporaneously and why it surfaces only under supply stress. But a regressor whose contribution is confined to one episode, identifiable only ex post, has no operational forecasting value. The honest reading is that HAR-X-QR captures a sample-specific covariation between the ENSO state and the cocoa supply crisis, rather than a replicable structural climate channel.

6.3 Forecast Instability and the Crisis Episode

The pattern just described is not an isolated peculiarity of cocoa. It is a textbook instance of the forecast-instability phenomenon documented in the financial-econometrics literature. Paye and Timmermann (2006) show that the discrepancy between strong in-sample predictability and weak out-of-sample performance, repeatedly observed across return-prediction studies, is best explained by structural instability in the underlying predictive relationship: predictability is real but concentrated in identifiable subperiods rather than uniform over time. Rossi (2020) argues that instability is the rule in macroeconomic and financial forecasting, not the exception, and that diagnostic procedures aimed at locating the periods in which predictability is concentrated should now be considered part of standard practice.

The expanding-window diagnostic in Section 5.5 is designed to operate within exactly that tradition. It imposes no break date, partitions no sample *ex post*, and traces the cumulative loss difference recursively from the first feasible forecast onward. The diagnostic identifies two episodes rather than one. The 1997–98 El Niño — the strongest on record at that time — produced a temporary gain at the median and $\tau = 0.75$ for long horizons that subsequently dissipated over the 2000s, reaching near zero well before the 2022–2024 crisis and failing to reappear in the intervening ENSO cycles. That dissipation is itself evidence of regime-conditional behaviour: ENSO produced a real but transient predictive gain when the cocoa supply system was under stress, and carried no incremental information in the years that followed. The finding that the cumulative ENSO gain at the twelve-month upper tail is negative — the augmentation harmful on net — through the three decades preceding 2022, and that the entire full-sample improvement accumulates within the final two years, is diagnostically clean: both partitions emerge from the data, not from the researcher. Poon and Granger (2003) warn that in-sample parameter stability is no guarantee of out-of-sample stability, and that time variation in parameter estimates is the critical issue in forecast evaluation; the expanding-window construction addresses this warning directly.

The Model Confidence Set complements rather than contradicts this picture. Hansen et al. (2011) stress that the MCS acknowledges the limitations of the data: informative samples yield small confidence sets, uninformative samples include many models. The wide long-horizon confidence sets — all models except historical volatility share $\mathcal{M}^*$ from three months onward — are therefore a statement about the limits of the data: averaged over the full sample, the loss differentials at long horizons are small relative to their sampling variation, and the MCS cannot, by construction, separate a stable channel from a single episode that dominates the average loss. Read together, the MCS result and the expanding-window diagnostic are entirely consistent: the former says the full-sample averages barely discriminate among models at long horizons; the latter

says that what little discrimination exists is shaped by one specific subperiod.

6.4 Coverage, Calibration, and the Discrimination Tradeoff

Two coverage results in Section 5.2 appear, on a first reading, to be in tension with the quantile-loss rankings. The HAR-QR violation rate at the twelve-month upper quantile rises to 14.63% against a nominal 5%, and the HAR-X-QR violation rate rises further to 15.89%, yet at that horizon HAR-X-QR achieves a modestly lower quantile loss than HAR-QR. Christoffersen (1998) provides the relevant distinction: unconditional coverage and conditional coverage are separate properties, and a forecast can satisfy the former while violating the latter through clustered exceedances. On the non-overlapping subsequences used for testing, the LR_{uc} test rejects unconditional coverage at six months for both models, and for HAR-X-QR additionally at one and three months (Table 15); at twelve months the subsequence is too short for a powerful test, so the 14–16% empirical violation rates are the more informative summary. The concentration of violations in the 2022–2024 episode is directly visible in Figure 7 (panel b2), where realized volatility remains above the $\tau = 0.95$ boundary for an extended stretch before the forecast distribution adjusts, so the test statistics understate the true calibration problem.

The apparent contradiction between higher violations and lower quantile loss is resolved by recognising that the two metrics measure orthogonal properties of a distributional forecast. Dimitriadis and Puke (2026) formalise this distinction in a score-decomposition framework: average loss under a strictly consistent scoring function rewards *discrimination*, the ability to distinguish high-risk from low-risk states, whereas backtest hit rates assess *calibration*, the ability to match nominal coverage. The two are not jointly maximised. The ENSO coefficient in HAR-X-QR raises the upper-quantile forecast during El Niño episodes, which reduces the magnitude of large exceedances when realised volatility spikes, and lowers it during La Niña, which produces more small exceedances in otherwise calm periods. The asymmetric check function penalises a small number of large under-predictions more heavily than a larger number of small ones, so a model that re-allocates its boundary toward the conditions that produce the most damaging exceedances can win on quantile loss while losing on average violation rate. At the twelve-month horizon this is exactly what happens: HAR-X-QR raises the boundary during the crisis enough to reduce the largest exceedances, gaining 3.3% in quantile loss at the cost of a higher violation rate. The qualification matters, however: at the one- to six-month horizons ENSO augmentation worsens loss and coverage *simultaneously*, so the discrimination–calibration tradeoff rationalises the twelve-month result only, not the augmentation in general.

Patton (2011) provides a complementary argument for prioritising the score over

the hit rate in this setting. Among the loss functions commonly used in volatility evaluation, only MSE and QLIKE are robust to noise in the volatility proxy; coverage diagnostics compare a binary indicator against the noisy proxy directly, and their informativeness deteriorates as proxy noise rises with the forecast horizon. The Yang-Zhang proxy becomes noisier as the averaging horizon extends; the quantile-loss ranking is, on Patton’s argument, more trustworthy at twelve months than the violation-rate comparison. Which property a user should prioritise depends on the use case: distributional ranking favours the score, nominal-coverage compliance favours the violation rate, and the two should be reported together.

6.5 Limitations and Future Research

Six limitations of the present specification deserve explicit mention. First, the HAR-QR coefficients are static within each estimation window. Gong et al. (2017) show that a HAR specification with time-varying sparsity, using normal-gamma autoregressive priors, substantially outperforms standard HAR for agricultural commodity futures; a time-varying HAR-QR with smoothly evolving quantile coefficients is a natural next step and would be expected to reduce the coverage degradation observed at long horizons. Second, the model has no explicit regime structure. Luo et al. (2022) demonstrate that infinite-Hidden-Markov regime switching delivers substantial gains over benchmark HAR for agricultural commodities precisely when macroeconomic uncertainty is high; the 2022–2024 episode is exactly the kind of high-uncertainty regime that an IHM-HAR-QR would be designed to model directly. Third, the model conditions on volatility lags but not on realised jumps, skewness, or higher moments; Andersen et al. (2007) and Bonato et al. (2024) both show that these components carry forecast-relevant information beyond what HAR lags capture, and the upper-tail coverage failures at long horizons are a plausible signature of unmodelled jumps in the crisis period.

Fourth, the climate exposure is summarised by a single index. The Indian Ocean Dipole and the Atlantic Meridional Mode also influence West African rainfall patterns and could disentangle Pacific from Atlantic teleconnections in a multi-index extension. Fifth, the analysis uses ICE London front-month rolling futures denominated in GBP only; ICE New York contracts, basis dynamics between the two markets, and contract-structure effects are not modelled, and a comparative two-exchange analysis would provide a natural robustness extension. Sixth, the estimation window length is fixed at 2,520 trading days. Bouri et al. (2021) report that the predictive value of ENSO on oil volatility emerges most strongly under estimation windows of four to six years; the ten-year window used here may dilute regime-specific coefficients and dampen climate-driven episodes. An adaptive-window design that lets the data choose the relevant

historical horizon for each forecast is a methodological extension worth pursuing.

Beyond these direct extensions, the broader research programme suggested by the cocoa results points in three directions. A jump-augmented HAR-J-QR specification would address the upper-tail coverage problem at the source. A regime-switching IHM-HAR-QR would model directly what the present specification absorbs through a single ENSO coefficient. A multi-predictor extension combining the Niño 3.4 index with realised inventory data, shipping rates, or other proxies of physical supply tightness would test whether the ENSO gain identified here can be disentangled from the broader regime indicators that coincide with it. Each of these directions would require both the data infrastructure and the methodological development that lie outside the scope of a single master's thesis; together they define the natural research agenda that follows from the findings reported in Section 5.

The overall picture is consistent. HAR-based quantile regression provides a statistically significant improvement over the standard volatility-forecasting benchmarks for cocoa at the daily-to-monthly horizons, where it holds the top rank in an upper-tail Model Confidence Set from which all three benchmarks are excluded. At the six- and twelve-month horizons that motivated the procurement perspective, simpler quantile constructions anchored to the unconditional distribution prove more robust, and no model achieves acceptable upper-tail calibration through the crisis. The ENSO augmentation produces no robust gain at any horizon: its modest full-sample twelve-month improvement reflects a single severe episode rather than a replicable climate channel. The combination of a clearly delimited RQ1 result and a negative RQ2 result positions the thesis as a methodological contribution to the volatility-forecasting literature on agricultural commodities, while flagging precisely where the limits of what daily-frequency data and three decades of evaluation can reveal about a slow-moving climate signal in fact lie.

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Appendices

A Supplementary Tables

This appendix collects reference material and detailed estimates that support the main text but are not essential to the line of argument: the data description (Table 10), the unit root tests (Table 11), the Mincer-Zarnowitz validation of the Yang-Zhang proxy against intraday realized volatility (Table 12), the full five-model quantile-loss comparison across quantile levels (Table 13), the full coefficient estimates of the HAR-QR and HAR-X-QR models (Table 14), the Christoffersen (1998) conditional coverage test results (Table 15), and the two corroborating RQ2 robustness diagnostics — the ENSO lag-decay table (Table 16) and the cross-commodity comparison (Table 17).

Table 10: Summary of Cocoa Futures Data

Characteristic	Value
Exchange	ICE Futures Europe
Contract	LCC (Cocoa Futures)
Contract size	10 metric tonnes
Price quotation	GBP per metric tonne
Continuous series	LCCc1 (front-month rolling)
Data source	Refinitiv Workspace
Sample period (daily)	January 2, 1985 – December 15, 2025
Number of observations (daily)	10,376 trading days
Sample period (5-minute)	February 3, 2025 – January 12, 2026
Number of observations (5-minute)	20,791 intraday observations
Variables (daily)	Open, High, Low, Close, Volume
Variables (5-minute)	Open, High, Low, Close, Volume, Bid, Ask

Table 11: Unit root tests for stationarity

Series	N	ADF		PP	
		Statistic	p -value	Statistic	p -value
Log returns	10,375	−72.06	< 0.001	−98.89	< 0.001
Yang-Zhang volatility	10,375	−6.27	< 0.001	−77.58	< 0.001
Niño 3.4 anomaly	787	−6.85	< 0.001	−6.46	< 0.001

ADF = Augmented Dickey–Fuller test (Dickey and Fuller 1979); PP = Phillips–Perron test (Phillips and Perron 1988). Both tests include a constant. ADF lag length selected by AIC. The null hypothesis is a unit root; rejection indicates stationarity. The Niño 3.4 series is tested on the full available NOAA history ($N = 787$ monthly observations, 1960–2025) rather than the regression subsample (1985–2025, $N = 492$), as unit root tests have higher power with longer samples.

Table 12: Mincer-Zarnowitz regression: Yang-Zhang against intraday realized volatility

	$\hat{\alpha}$	$\hat{\beta}$	R^2	N	$F_{\alpha=0, \beta=1}$	p -value
Daily	0.0127*** (0.0011)	0.4589*** (0.0443)	0.464	196	75.1	< 0.001
22-day rolling	0.0022*** (0.0006)	0.8355*** (0.0247)	0.847	175	140.9	< 0.001

Mincer-Zarnowitz regression $RV_t = \alpha + \beta \hat{\sigma}_t^{YZ} + \varepsilon_t$ estimated with HC1 standard errors (in parentheses). The F -statistic tests the joint hypothesis $\alpha = 0, \beta = 1$ (unbiased proxy). RV is computed from five-minute log returns; $\hat{\sigma}^{YZ}$ is the Yang-Zhang estimator from daily OHLC prices.

The 22-day row uses rolling 22-day averages of both series, aligning with the HAR monthly component. The lower $\hat{\beta}$ at the daily frequency reflects the different treatment of the overnight component: Yang-Zhang captures the close-to-open gap while intraday RV does not. Sample: 3 February 2025 – 12 January 2026. *** $p < 0.001$.

Table 13: Quantile loss by model and horizon across quantile levels

Model	1d	1w	1m	3m	6m	12m
<i>Panel A: $\tau = 0.50$</i>						
Historical Vol	0.002731	0.001906	0.001530	0.001499	0.001531	0.001674
GARCH(1,1)	0.003313	0.002518	0.002201	0.002105	0.002113	0.002155
HAR-OLS	0.002467	0.001637	0.001264	0.001213	0.001291	0.001387
HAR-QR	0.002470	0.001639	0.001267	0.001201	0.001304	0.001381
HAR-X-QR	0.002470	0.001645	0.001275	0.001219	0.001317	0.001358
<i>Panel B: $\tau = 0.75$</i>						
Historical Vol	0.002672	0.001790	0.001444	0.001391	0.001480	0.001706
GARCH(1,1)	0.003099	0.002176	0.001746	0.001639	0.001633	0.001675
HAR-OLS	0.002408	0.001557	0.001184	0.001118	0.001172	0.001290
HAR-QR	0.002396	0.001523	0.001159	0.001109	0.001147	0.001307
HAR-X-QR	0.002397	0.001528	0.001175	0.001146	0.001187	0.001395
<i>Panel C: $\tau = 0.95$</i>						
Historical Vol	0.001281	0.000830	0.000691	0.000809	0.001033	0.001462
GARCH(1,1)	0.001283	0.000828	0.000609	0.000629	0.000668	0.000736
HAR-OLS	0.001229	0.000746	0.000536	0.000558	0.000608	0.000786
HAR-QR	0.001159	0.000682	0.000500	0.000557	0.000704	0.001041
HAR-X-QR	0.001160	0.000685	0.000521	0.000580	0.000711	0.001007

Quantile loss (Equation (18)); lower values indicate better forecasts. Bold: best model at each horizon within a panel. The $\tau = 0.95$ panel reproduces Table 4 from the main text so the three panels can be read together. HAR-OLS quantiles are pseudo-quantiles from its in-sample residual distribution; HAR-QR and HAR-X-QR estimate the quantile directly.

Table 14: Full-sample HAR-QR and HAR-X-QR coefficient estimates

Horizon	Model	β_0	β_d	β_w	β_m	β_{ENSO}
<i>Panel: $\tau = 0.50$</i>						
1 day	HAR-QR	0.0014	0.1231	0.1721	0.4950	—
	HAR-X-QR	0.0015	0.1195	0.1732	0.4900	-0.0001
1 week	HAR-QR	0.0023	0.0729	0.1317	0.5745	—
	HAR-X-QR	0.0024	0.0717	0.1341	0.5671	-0.0001
1 month	HAR-QR	0.0034	0.0329	0.1264	0.5669	—
	HAR-X-QR	0.0034	0.0345	0.1210	0.5650	-0.0002
3 months	HAR-QR	0.0044	0.0186	0.0684	0.5670	—
	HAR-X-QR	0.0045	0.0181	0.0718	0.5553	-0.0003
6 months	HAR-QR	0.0060	0.0199	0.0489	0.4850	—
	HAR-X-QR	0.0061	0.0199	0.0442	0.4761	-0.0004
12 months	HAR-QR	0.0078	0.0144	0.0400	0.3626	—
	HAR-X-QR	0.0078	0.0131	0.0399	0.3606	-0.0002
<i>Panel: $\tau = 0.75$</i>						
1 day	HAR-QR	0.0026	0.1742	0.2477	0.5819	—
	HAR-X-QR	0.0027	0.1772	0.2471	0.5689	-0.0001
1 week	HAR-QR	0.0026	0.0975	0.1989	0.6606	—
	HAR-X-QR	0.0028	0.0985	0.1934	0.6566	-0.0000
1 month	HAR-QR	0.0039	0.0507	0.1402	0.6533	—
	HAR-X-QR	0.0039	0.0468	0.1341	0.6599	-0.0002
3 months	HAR-QR	0.0054	0.0296	0.0559	0.6377	—
	HAR-X-QR	0.0055	0.0280	0.0547	0.6317	-0.0003
6 months	HAR-QR	0.0070	0.0209	0.0463	0.5485	—
	HAR-X-QR	0.0070	0.0192	0.0488	0.5419	-0.0003
12 months	HAR-QR	0.0090	0.0133	0.0333	0.4250	—
	HAR-X-QR	0.0090	0.0136	0.0334	0.4249	-0.0000
<i>Panel: $\tau = 0.95$</i>						
1 day	HAR-QR	0.0055	0.3740	0.4437	0.7203	—
	HAR-X-QR	0.0055	0.3787	0.4366	0.7209	-0.0002
1 week	HAR-QR	0.0042	0.2051	0.3034	0.7817	—
	HAR-X-QR	0.0041	0.2057	0.3074	0.7814	-0.0002
1 month	HAR-QR	0.0050	0.0559	0.1349	0.9099	—
	HAR-X-QR	0.0050	0.0520	0.1563	0.8886	-0.0001
3 months	HAR-QR	0.0068	0.0376	0.2375	0.6378	—
	HAR-X-QR	0.0068	0.0303	0.2437	0.6368	-0.0001
6 months	HAR-QR	0.0076	0.0261	0.1359	0.6769	—
	HAR-X-QR	0.0079	0.0205	0.1526	0.6602	0.0003
12 months	HAR-QR	0.0104	0.0225	0.1217	0.4857	—
	HAR-X-QR	0.0134	0.0082	0.1092	0.4131	0.0026

Note: Quantile regression coefficients estimated once on the full sample for each horizon and quantile τ . β_0 is the intercept; β_d , β_w , β_m are the daily, weekly and monthly HAR components; β_{ENSO} is the Niño 3.4 loading (HAR-X-QR only). The table is descriptive: it reports the sign and magnitude of the loadings only. Significance tests are omitted because the forecast targets are overlapping h -day-ahead averages, which violates the independence assumption underlying the standard errors. The forecasting results in Section 5 use rolling-window re-estimation; these full-sample estimates are reported for reference only.

Table 15: Christoffersen (1998) conditional coverage tests, $\tau = 0.95$

Horizon	Model	N	N_{no}	VR (%)	LR_{uc}	LR_{ind}	LR_{cc}
1 day	HAR-QR	7737	7737	5.17	0.466	7.032***	7.498**
1 day	HAR-X-QR	7737	7737	5.14	0.335	6.256**	6.592**
1 week	HAR-QR	7733	1547	5.39	0.586	0.045	0.631
1 week	HAR-X-QR	7733	1547	5.41	0.179	0.448	0.626
1 month	HAR-QR	7716	351	6.35	1.103	1.682	2.785
1 month	HAR-X-QR	7716	351	7.14	5.591**	3.149*	8.741**
3 months	HAR-QR	7672	117	8.15	0.75	2.88*	3.63
3 months	HAR-X-QR	7672	117	9.68	5.29**	5.479**	10.769***
6 months	HAR-QR	7570	58	11.45	6.523**	0.81	7.333**
6 months	HAR-X-QR	7570	58	14.15	11.514***	3.586*	15.1***
12 months	HAR-QR	7306	28	14.63	1.471	—	—
12 months	HAR-X-QR	7306	28	15.89	3.461*	0.754	4.216

LR_{uc} : unconditional coverage test (χ_1^2); LR_{ind} : independence of violations (χ_1^2); LR_{cc} : conditional coverage (χ_2^2). VR: empirical violation rate (%) over all N daily forecasts; nominal target = 5%.

Because daily forecasts at $h > 1$ target overlapping h -day windows, consecutive violations are dependent by construction; the LR tests are therefore computed on the non-overlapping subsequence of every h -th forecast (N_{no} observations). Cells marked — are not computable (no violations or too few observations in the non-overlapping subsequence). * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 16: Lag-decay diagnostic: quantile loss ($\tau = 0.95$) by ENSO lag

Horizon	ENSO lag (months)					HAR-QR (no ENSO)
	0	1	3	6	12	
1m	0.000521	0.000521	0.000507	0.000505	0.000507	0.000500
6m	0.000712	0.000691	0.000705	0.000713	0.000787	0.000704
12m	0.001007	0.001004	0.001091	0.001140	0.001046	0.001041

Lower values indicate better forecasts. Bold: best lag for each horizon. Lag k : Niño 3.4 from k calendar months prior to the forecast date; construction is fully causal. Final column: HAR-QR baseline on the identical sample. Where ENSO helps at the twelve-month upper tail it does so only at lag 0 or lag 1; the agronomically motivated lags of three and six months are worse than the no-ENSO baseline.

Table 17: Cross-commodity ENSO gain (ΔQL , $\tau = 0.95$) at 6- and 12-month horizons

Horizon	Cocoa	Coffee	Sugar
6m	+0.000007	+0.000015	−0.000010
12m	−0.000034	+0.000079	−0.000007

$\Delta QL = QL_{HAR-X-QR} - QL_{HAR-QR}$; negative values indicate improvement from ENSO augmentation. Coffee: ICE KC=F, $N = 3,538$ pairs at 12m. Sugar: ICE SB=F, $N = 3,500$ pairs at 12m. Cocoa: $N = 7,306$. No commodity shows a material gain; coffee worsens at both horizons, sugar's gain is negligible.